MEAN CURVATURE FLOW IN GENERAL AMBIENT MANIFOLDS

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Abstract. We establish short-time existence of mean curvature flow starting from a complete immersed submanifold with bounded second fundamental form in a complete ambient manifold. The ambient curvature and its first two covariant derivatives are assumed to be bounded, and no positive lower bound for the injectivity radius is required. The existence time and the curvature bound depend only on these bounds and the dimensions.

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1. Introduction

Let $F_0 : M^n \to (N^{n+m}, h)$ be a smooth immersion with complete induced metric. A smooth one-parameter family of immersions $F(\cdot, t) : M \to N$ is said to evolve by mean curvature flow if

$$\partial_t F = \mathbf{H}, \qquad F|_{t=0} = F_0, \tag{1.1}$$

where $\mathbf{H}$ denotes the mean curvature vector of the immersion $F(\cdot, t)$.

For a compact initial submanifold, short-time existence follows from the standard theory of quasilinear parabolic systems. The situation is more subtle for complete noncompact initial submanifolds. In the Euclidean setting, Ecker and Huisken [4] established existence results for graphical hypersurfaces, while Wang [13] developed the corresponding theory in arbitrary codimension. Adapting the approach of Koch and Lamm [9], the author [5] established well-posedness results for mean curvature flow of hypersurfaces. For ambient manifolds of bounded geometry, Chen and Yin [2] proved uniqueness and established a pseudolocality theorem for mean curvature flow. Smoczyk [11, Section 3.1, p. 249] pointed out that the short-time existence and uniqueness problem for complete noncompact initial

Date: November 9, 2026.
2020 *Mathematics Subject Classification.* 53E10.
Key words and phrases. Mean curvature flows, short-time existence, uniqueness.

submanifolds is open in general, despite a number of important results under additional geometric assumptions.

The purpose of the paper is to establish short-time existence without imposing a uniform positive lower bound on the injectivity radius of the ambient manifold. Our main result is the following.

Theorem 1.1. *Let (N^{n+m}, h) be a smooth complete Riemannian manifold, and let $F_0 : M^n \to N$ be a smooth immersion with complete induced metric. Suppose*

$$\begin{aligned} &\sup_N |\operatorname{Rm}_N| \le K_0, \qquad \sup_N |\nabla^N \operatorname{Rm}_N| \le K_1, \\ &\sup_N |(\nabla^N)^2 \operatorname{Rm}_N| \le K_2, \qquad \sup_M |A_0| \le \Lambda. \end{aligned} \tag{1.2}$$

Then there exists $T = T(n, m, K_0, K_1, K_2, \Lambda) > 0$ and a smooth family of immersions $F_t : M \to N$, $0 \le t \le T$, such that (1.1) holds and every induced metric is complete. Moreover,

$$\sup_{M\times[0,T]} |A_t|^2 \le C(n, m)\big(\Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2}\big). \tag{1.3}$$

If $\kappa := \Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2} > 0$, one may take $T = c(n, m)/\kappa$. If $\kappa = 0$, the stationary flow $F_t = F_0$ exists for all time.

Theorem 1.1 shows that both the existence time and the curvature estimate depend only on the dimension, the ambient curvature bounds, and the initial bound for the second fundamental form, and are independent of the injectivity radius of the ambient manifold. In particular, the theorem applies uniformly in the collapsing regime.

Since no uniform lower bound on the injectivity radius of the ambient manifold is assumed, one cannot in general expect a uniform lower bound for the injectivity radius of the induced metric on the submanifold. In particular, the usual Calderón–Zygmund estimates under bounded geometry are not directly available. A main difficulty in the proof of Theorem 1.1 is therefore to establish a Calderón–Zygmund estimate for sections of the pull-back bundle $E = F_0^*TN$ whose constants are independent of the injectivity radius. The weighted spaces used in the construction are motivated by [9, 5]. We obtain a solution of the mean curvature–DeTurck equation by the Banach fixed-point theorem. We compute the full nonlinear remainder using integral equations for Jacobi fields and prove a uniform two-section estimate directly. The second ambient curvature derivative controls the comparison of the transported first curvature derivative along two radial geodesics.

The heat kernel used in the argument is that of the fixed connection Laplacian Δ^E. The full linearisation also contains first-order coefficients involving A_0. Although A_0 is smooth locally, its derivatives need not be uniformly bounded under our assumptions. We therefore place every lower-order linear term on the right-hand side and estimate it by the heat potential of Δ^E. No second-derivative or $C^{2,\alpha}$ heat-kernel estimate for the full linearisation is used. Uniform local parabolic estimates then give a bound for the second fundamental form and the DeTurck vector field, which allows the gauge to be removed on the complete manifold. A related strategy has recently been successfully implemented for Ricci flow in the collapsing setting; see [1].

The same ambient curvature assumptions also give the following uniqueness statement.

Theorem 1.2. *Let (N^{n+m}, h) be a complete Riemannian manifold satisfying*

$$\sup_N \big(|\operatorname{Rm}_N| + |\nabla^N \operatorname{Rm}_N| + |(\nabla^N)^2 \operatorname{Rm}_N|\big) < \infty.$$

Let $F_0 : M^n \to N$ be a smooth immersion such that $g_0 = F_0^* h$ is complete. Suppose that $F_1, F_2 : M \times [0,T] \to N$ are smooth mean curvature flows with the same initial immersion F_0, and that each induced metric $F_i(t)^* h$ is complete and

$$\sup_{M\times[0,T]} \big(|A_{F_1}| + |A_{F_2}|\big) < \infty.$$

Then

$$F_1(p,t) = F_2(p,t) \qquad \forall (p,t) \in M \times [0,T].$$

Disclosure of AI use. The author acknowledges the use of ChatGPT as an auxiliary tool in revising the exposition and developing parts of the arguments. The author is responsible for reviewing the AI-assisted revisions and verifying all mathematical claims before submission.

Acknowledgments. The author is supported by the Project of Stable Support for Youth Team in Basic Research Field, Chinese Academy of Sciences (YSBR-001).

2. Preliminaries

Let (N^{n+m}, h) be a smooth complete Riemannian manifold and let

$$F_0 : (M^n, g_0) \longrightarrow (N^{n+m}, h)$$

be a smooth isometric immersion. The metric $g_0 = F_0^* h$ is assumed complete. We write A_0 and $\mathbf{H}_0$ for the second fundamental form and the mean-curvature vector of F_0. Replace h by $\widetilde{h} = \rho^{-2} h$ and t by $\widetilde{t} = \rho^{-2} t$. Then

$$|\widetilde{A}_0| = \rho |A_0|, \qquad |\operatorname{Rm}_{\widetilde{h}}| = \rho^2 |\operatorname{Rm}_h|,$$
$$|\widetilde{\nabla} \operatorname{Rm}_{\widetilde{h}}| = \rho^3 |\nabla \operatorname{Rm}_h|, \qquad |\widetilde{\nabla}^2 \operatorname{Rm}_{\widetilde{h}}| = \rho^4 |\nabla^2 \operatorname{Rm}_h|.$$

If $\kappa = \Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2} > 0$, choose $\rho = \kappa^{-1/2}$. Then $\rho\Lambda, \rho^2 K_0, \rho^3 K_1, \rho^4 K_2 \le 1$. Therefore, it suffices to prove the main theorem under the normalised assumptions

$$|A_0| \le 1, \qquad |\operatorname{Rm}_N| \le 1, \qquad |\nabla^N \operatorname{Rm}_N| \le 1, \qquad |(\nabla^N)^2 \operatorname{Rm}_N| \le 1, \tag{2.1}$$

for a time $T_* = T_*(n,m) > 0$. Scaling back gives $T = \rho^2 T_*$. Unless otherwise stated, constants in the normalized geometric estimates depend only on n and m.

2.1. Intrinsic and bundle curvature.

Let $E := F_0^* TN$ and equip E with the pull-back metric and connection ∇^E. The Gauss equation is

$$\begin{aligned}\operatorname{Rm}_M(X,Y,Z,W) = {} & \operatorname{Rm}_N(dF_0 X, dF_0 Y, dF_0 Z, dF_0 W) \\ & + \langle A_0(X,W), A_0(Y,Z)\rangle - \langle A_0(X,Z), A_0(Y,W)\rangle.\end{aligned} \tag{2.2}$$

Thus

$$|\operatorname{Rm}_M| \le C(n)(|\operatorname{Rm}_N| + |A_0|^2) \le C(n). \tag{2.3}$$

In particular, $\operatorname{Ric}_M \ge -C(n) g_0$.

The curvature of E is

$$R^E(X,Y)s = R^N(dF_0 X, dF_0 Y)s. \tag{2.4}$$

Differentiating (2.4), with the induced connections on all tensor factors, gives

$$(\nabla_X R^E)(Y,Z)s = (\nabla^N_{dF_0X} R^N)(dF_0Y, dF_0Z)s \\ + R^N(A_0(X,Y), dF_0Z)s + R^N(dF_0Y, A_0(X,Z))s. \tag{2.5}$$

Therefore

$$|R^E| + |\nabla^E R^E| \le C(n,m). \tag{2.6}$$

Write $V_x(r) = \mathrm{Vol}_{g_0}(B_{g_0}(x,r))$. By Bishop–Gromov comparison and (2.3), for $0 < r \le R \le 1$,

$$\frac{V_x(R)}{V_x(r)} \le C(n)\left(\frac{R}{r}\right)^n. \tag{2.7}$$

For a section U of E, define

$$\mathcal{E}(U)(x) := \mathrm{Exp}^N_{F_0(x)} U(x).$$

Lemma 2.1. *There are $r_* = r_*(n,m) > 0$, $\varepsilon_* = \varepsilon_*(n,m) > 0$, and $C = C(n,m)$ with the following property. If*

$$\|U\|_{L^\infty} \le r_*, \qquad \|\nabla^E U\|_{L^\infty} \le \varepsilon_*, \tag{2.8}$$

then $F_U := \mathcal{E}(U)$ is an immersion and

$$(1 - C(\varepsilon_* + r_*^2))g_0 \le F_U^* h \le (1 + C(\varepsilon_* + r_*^2))g_0. \tag{2.9}$$

The first and second covariant derivatives of Exp^N, measured using the horizontal–vertical splitting of TTN induced by h, are uniformly bounded on $\{|v| \le r_*\} \subset TN$.

Proof. Let $\gamma(s) = \mathrm{Exp}_{F_0(x)}(sU(x))$, $0 \le s \le 1$, and let P_s denote parallel transport along γ. If $x(a)$ is a curve in M with $x'(0) = X$, differentiating $\mathrm{Exp}_{F_0(x(a))}(sU(x(a)))$ in a gives a Jacobi field J along γ. At $s = 0$,

$$J(0) = dF_0(X), \qquad \nabla_s J(0) = \nabla^E_X U.$$

After parallel transport back to $T_{F_0(x)}N$, the Jacobi equation is

$$J''(s) = -R^N(J(s), U)U.$$

Integrating this equation twice and using $|\,\mathrm{Rm}_N\,| \le 1$ gives

$$\big|P_s^{-1}J(s) - dF_0(X) - s\nabla^E_X U\big| \le C|U|^2\big(|X| + |\nabla^E_X U|\big). \tag{2.10}$$

At $s = 1$, $J(1) = dF_U(X)$. Hence

$$|dF_U(X) - P_1 dF_0(X)| \le C\big(|\nabla^E_X U| + |U|^2|X|\big).$$

Choosing r_* and ε_* sufficiently small proves that dF_U is injective and gives (2.9).

Differentiating the Jacobi equation once with respect to the initial data gives an inhomogeneous Jacobi equation. Its coefficients and forcing terms are controlled by $|\,\mathrm{Rm}_N\,| + |\nabla^N \mathrm{Rm}_N\,|$ and the first Jacobi-field estimates. Integration on $[0,1]$ therefore gives the asserted second-derivative bounds. This argument uses the exponential map on a ball in each tangent space; it does not require the map to be globally injective on that ball. □

2.2. **Linearization of the mean curvature–DeTurck flow.** For an immersion $F : M \to N$, let $g(F) = F^*h$ and define

$$W(F)^k = g(F)^{ij}\big(\Gamma(g(F))^k_{ij} - \Gamma(g_0)^k_{ij}\big). \tag{2.11}$$

In local coordinates, the mean curvature vector is given by

$$\mathbf{H}(F)^\alpha = g(F)^{ij}\left(\partial_i\partial_j F^\alpha - \Gamma(g(F))^k_{ij}\partial_k F^\alpha + \overline{\Gamma}^\alpha_{\beta\gamma}(F)\partial_i F^\beta \partial_j F^\gamma\right). \tag{2.12}$$

Combining (2.11) and (2.12), we get

$$\begin{aligned}\mathbf{H}(F) + dF(W(F)) &= g(F)^{ij}\left(\partial_i\partial_j F^\alpha - \Gamma(g_0)^k_{ij}\partial_k F^\alpha + \overline{\Gamma}^\alpha_{\beta\gamma}(F)\partial_i F^\beta \partial_j F^\gamma\right)\partial_\alpha \\ &= \operatorname{tr}_{g(F)} D^{g_0,h}dF.\end{aligned} \tag{2.13}$$

Its principal symbol is $g(F)^{ij}\xi_i\xi_j \operatorname{Id}_{F^*TN}$, so it is strongly parabolic as long as $g(F)$ remains uniformly equivalent to g_0.

We solve

$$\partial_t\widehat{F} = \mathbf{H}(\widehat{F}) + d\widehat{F}(W(\widehat{F})), \qquad \widehat{F}(\cdot,0) = F_0. \tag{2.14}$$

Taking normal projections immediately yields

$$(\partial_t\widehat{F})^\perp = \mathbf{H}(\widehat{F}).$$

Definition 2.2. A smooth family of immersions $\widehat{F} : M \times [0,T] \to N$ is called a mean curvature–DeTurck flow if it satisfies (2.14).

Write $\widehat{F} = F_U = \mathcal{E}(U)$. Let $P_U : E_x \to T_{F_U(x)}N$ be parallel transport along $s \mapsto \operatorname{Exp}_{F_0(x)}(sU(x))$, and define the vertical differential

$$\mathcal{V}_U := P_U^{-1} \circ d_v \operatorname{Exp}_{(F_0,U)} : E_x \longrightarrow E_x. \tag{2.15}$$

By Lemma 2.1, $\mathcal{V}_U$ is invertible if U satisfies (2.8) and $\mathcal{V}_0 = \mathrm{Id}$. Since the base point $F_0(x)$ is independent of time,

$$P_U^{-1}\partial_t F_U = \mathcal{V}_U(\partial_t U).$$

Consequently, (2.14) is equivalent to

$$\partial_t U = \mathcal{P}(U) := \mathcal{V}_U^{-1}P_U^{-1}\left\{\operatorname{tr}_{g(F_U)} D^{g_0,h}dF_U\right\}. \tag{2.16}$$

Let

$$\Delta^E U = g_0^{ij}\left(\nabla^E_i\nabla^E_j U - \Gamma(g_0)^k_{ij}\nabla^E_k U\right)$$

be the rough Laplacian of E.

Lemma 2.3. *Suppose that the normalised bounds* (2.1) *hold. For* $|U| + |\nabla^E U| \le \varepsilon_*$, *the exact gauged equation has the quasilinear form*

$$\partial_t U = g(F_U)^{ij}(\nabla^{E,2}U)_{ij} + b(x, U, \nabla^E U). \tag{2.17}$$

The coefficients are smooth in their finite-dimensional jet variables and locally smooth in x. *They satisfy, with* $z = |U| + |\nabla^E U|$,

$$|g(F_U)^{-1} - g_0^{-1}| \le C\big(|\nabla^E U| + |U|^2\big), \tag{2.18}$$

$$|b(x, U, \nabla^E U) - \mathbf{H}_0| \le Cz, \tag{2.19}$$

and $g(F_U)$ *is uniformly equivalent to* g_0. *In particular, the equation can be written as*

$$\partial_t U = \mathbf{H}_0 + L_0 U + \mathcal{Q}(U), \tag{2.20}$$

where

$$L_0U = \Delta^E U + \sum_i R^N(U, e_i)e_i - 2\sum_{i,j}\langle \nabla^E_{X_i} U, e_j\rangle A_{ij},$$

and the full nonlinear remainder satisfies the uniform one-section estimate

$$|\mathcal{Q}(U)| \le Cz\big(z + |\nabla^{E,2}U|\big). \tag{2.21}$$

Here $\{X_i\}$ is a g_0-orthonormal frame, $e_i = dF_0(X_i)$, and $A_{ij} = A_0(X_i, X_j)$. All constants depend only on n, m. The single-section bounds in this lemma use only the bounds for A_0, Rm_N, and $\nabla^N \mathrm{Rm}_N$.

Proof. We first obtain bounds for the exact coefficients. All fields in the following argument are parallel transported back to E_x when compared. Put $F_s = \mathrm{Exp}_{F_0}(sU)$ and, in a g_0-normal frame at the fixed point, set

$$T = \partial_s F_s, \qquad E_i = dF_s(X_i), \qquad K_i = D_s E_i, \qquad C_{ij} = D_i E_j - \Gamma(g_0)^k_{ij} E_k.$$

Write $Z_i = \nabla^E_{X_i} U$ and $H_{ij} = (\nabla^{E,2}U)_{ij}$. The Jacobi equation and its initial data give, for $0 \le s \le 1$,

$$E_i(s) = e_i + sZ_i + \int_0^s (s-\sigma)\widetilde{R}_U(\sigma)(U, E_i(\sigma))U\, d\sigma,$$

where $\widetilde{R}_U$ denotes the curvature tensor transported to E_x. Taking the supremum over $s \in [0,1]$ yields

$$\sup_s |E_i(s)| \le 1 + |Z_i| + C|U|^2 \sup_s |E_i(s)|.$$

Choose ε_* so that $C\varepsilon_*^2 \le 1/2$. Absorbing the last term and then differentiating the integral equation with respect to s gives

$$\sup_s |E_i(s)| \le C(1+z), \qquad K_i(s) = Z_i + \int_0^s \widetilde{R}_U(\sigma)(U, E_i(\sigma))U\, d\sigma.$$

Consequently,

$$\sup_s \big(|E_i(s) - e_i - sZ_i| + |K_i(s) - Z_i|\big) \le C|U|^2(1+z).$$

We spell out the commutations needed for the second spatial derivative. Since $D_sT = 0$ and $D_iT = K_i$, we have

$$D_sK_j = R^N(T, E_j)T, \qquad D_sC_{ij} = D_iK_j - \Gamma(g_0)^k_{ij}K_k + R^N(T, E_i)E_j.$$

Differentiate the first two terms of D_sC_{ij} once more. The reference connection is independent of s, and the curvature commutator gives

$$\begin{aligned} D_s\big(D_iK_j - \Gamma(g_0)^k_{ij}K_k\big) &= (\nabla^N_{E_i}R^N)(T, E_j)T + R^N(K_i, E_j)T \\ &\quad + R^N(T, C_{ij})T + R^N(T, E_j)K_i + R^N(T, E_i)K_j. \end{aligned}$$

The derivative of the last term is

$$D_s\big(R^N(T, E_i)E_j\big) = (\nabla^N_T R^N)(T, E_i)E_j + R^N(T, K_i)E_j + R^N(T, E_i)K_j.$$

Adding these two identities yields

$$\begin{aligned} D_s^2C_{ij} &= R^N(T, C_{ij})T + (\nabla^N_{E_i}R^N)(T, E_j)T + (\nabla^N_T R^N)(T, E_i)E_j \\ &\quad + R^N(K_i, E_j)T + R^N(T, E_j)K_i + 2R^N(T, E_i)K_j + R^N(T, K_i)E_j, \end{aligned} \tag{2.22}$$

with initial values

$$C_{ij}(0) = A_{ij}, \qquad D_s C_{ij}(0) = H_{ij} + R^N(U, e_i)e_j.$$

Equation (2.22) is linear in C_{ij}. Its other terms depend only on x, U, Z, and their norms are bounded by Cz^2. Indeed, $|T| = |U|$, the two curvature-derivative terms have two factors of U, and every remaining term contains U and K. Writing the twice-integrated equation for C_{ij} gives

$$\sup_s |C_{ij}(s)| \le C(1 + |H|) + C|U|^2 \sup_s |C_{ij}(s)| + Cz^2.$$

After the same absorption, $\sup_s |C_{ij}(s)| \le C(1 + |H|)$. Substituting this bound back into the integral equation therefore gives

$$\big|P_s^{-1} C_{ij}(s) - A_{ij} - s\big(H_{ij} + R^N(U, e_i)e_j\big)\big| \le C\big(z^2 + |U|^2|H|\big). \tag{2.23}$$

Only the bounds for R^N and $\nabla^N R^N$ enter this estimate; no comparison of $\nabla^N R^N$ along two different paths is made.

The part of $C_{ij}(s)$ depending on H_{ij} is precisely the Jacobi field with initial values $0, H_{ij}$. At $s = 1$ this part equals $P_1 \mathcal{V}_U H_{ij}$. Multiplying by $\mathcal{V}_U^{-1} P_1^{-1}$ in (2.16) proves that the exact principal coefficient is $g(F_U)^{ij}$, as asserted in (2.17). The remaining coefficient b is obtained by setting $H = 0$ in this construction.

For completeness, the matrix cancellations used to estimate the lower coefficient can be written without Taylor notation. In the chosen frame, write $g(F_U) = I + a + q$, where $a_{ij} = \langle Z_i, e_j\rangle + \langle e_i, Z_j\rangle$. Expanding the inner product of $E_i(1) = e_i + Z_i + (E_i(1) - e_i - Z_i)$ and $E_j(1)$ gives $|q| \le Cz^2$. For $G = g(F_U)^{-1}$, the exact identities

$$G - I = -G(a + q), \qquad G - I + a = G(a + q)a - Gq$$

give $|G - I| \le C(|Z| + |U|^2)$ and $|G - I + a| \le Cz^2$. The vertical Jacobi equation similarly gives $|\mathcal{V}_U - I| \le C|U|^2$, and

$$\mathcal{V}_U^{-1} - I = -\mathcal{V}_U^{-1}(\mathcal{V}_U - I)$$

gives the same bound for its inverse. Substituting these identities into (2.23) leaves, at order one, precisely $\Delta^E U + g_0^{ij} R^N(U, e_i)e_j - a^{ij} A_{ij}$. Every remaining term is bounded by $Cz^2 + Cz|H|$. This proves (2.18), (2.19), and (2.21). The exact identity (2.38) below records each term of this remainder and will be used to compare two sections.

□

Lemma 2.4 (Uniform difference estimate for the full remainder)**.** *Assume the normalised bounds* (2.1)*, including* $|\nabla^{N,2} \mathrm{Rm}_N| \le 1$*. At a fixed point* $x \in M$*, put*

$$z_U = |U| + |\nabla^E U|, \qquad h_U = |\nabla^{E,2} U|, \qquad d_j = |\nabla^{E,j}(U - V)| \quad (0 \le j \le 2),$$

and define z_V, h_V *in the same way. There are dimensional constants* $\varepsilon_* > 0$ *and* C *such that, if* $z_U, z_V \le \varepsilon_*$*, then the full remainder in* (2.20) *satisfies*

$$|\mathcal{Q}(U) - \mathcal{Q}(V)| \le C(z_U + z_V)(d_0 + d_1 + d_2) + C(h_U + h_V)(d_0 + d_1). \tag{2.24}$$

Consequently, if

$$\mathcal{D}_0 = L_0 - \Delta^E, \qquad \mathcal{N}(U) = \mathcal{D}_0 U + \mathcal{Q}(U),$$

then, for $0 < T \le 1/4$ *and* $\|U\|_{\mathcal{X}_T}, \|V\|_{\mathcal{X}_T} \le \delta \le \varepsilon_*/2$*,*

$$\|\mathcal{N}(U) - \mathcal{N}(V)\|_{\mathcal{Y}_T} \le C(\sqrt{T} + \delta)\|U - V\|_{\mathcal{X}_T}. \tag{2.25}$$

Here $\mathcal{D}_0$ is treated as a first-order term on the right-hand side of the equation; no heat-kernel estimate for L_0 is used.

Proof. Fix x, choose a g_0-orthonormal frame at x, and put $q = F_0(x)$, $e_i = dF_0(X_i)$, $Z_i = \nabla^E_{X_i} U$, and $W_i = \nabla^E_{X_i} V$. All tensors along the radial geodesics below are parallel transported to the fixed space $E_x = T_qN$. Write

$$r_U = |U|, \quad r_V = |V|, \quad r_\Sigma = r_U + r_V, \quad z_\Sigma = z_U + z_V, \quad d = d_0 + d_1.$$

We choose ε_* small enough that all the Jacobi fields below have uniform bounds and all the indicated matrices are invertible. All constants in the proof are independent of x.

Step 1. Comparison after radial parallel transport. Let $u_\lambda = V + \lambda(U - V)$ and $\gamma_\lambda(s) = \operatorname{Exp}_q(su_\lambda)$, for $0 \le \lambda, s \le 1$. The variational field $Y_\lambda = \partial_\lambda \gamma_\lambda$ is a Jacobi field with initial data $Y_\lambda(0) = 0$ and $D_s Y_\lambda(0) = U - V$. After radial parallel transport, its integral equation is

$$Y_\lambda(s) = s(U - V) + \int_0^s (s - \sigma)\widetilde{R}_\lambda(\sigma)(u_\lambda, Y_\lambda(\sigma))u_\lambda \, d\sigma.$$

If $M_Y = \sup_{0<s\le 1} |Y_\lambda(s)|/s$, then $\int_0^s (s-\sigma)\sigma \, d\sigma = s^3/6$ gives

$$M_Y \le d_0 + Cr_\Sigma^2 M_Y/6.$$

Choosing ε_* so that $Cr_\Sigma^2/6 \le 1/2$ proves $M_Y \le 2d_0$. Differentiating the integral equation now gives

$$|Y_\lambda(s)| \le Csd_0, \qquad |D_s Y_\lambda(s)| \le Cd_0, \qquad |\partial_s \gamma_\lambda| = |u_\lambda| \le r_\Sigma.$$

Let $P_\lambda(s) : E_x \to T_{\gamma_\lambda(s)}N$ be radial parallel transport and set $\mathscr{A}_\lambda = P_\lambda^{-1} D_\lambda P_\lambda$. Commutation of the two covariant derivatives gives

$$\partial_s \mathscr{A}_\lambda = P_\lambda^{-1} R^N(\partial_s \gamma_\lambda, Y_\lambda) P_\lambda, \qquad \mathscr{A}_\lambda(0) = 0.$$

In particular,

$$|\mathscr{A}_\lambda(s)| \le Cr_\Sigma s^2 d_0.$$

For a tensor $\mathscr{T}$ of type $(1,k)$, write $\widetilde{\mathscr{T}}(a_1, \dots, a_k) = P_\lambda^{-1}\mathscr{T}(P_\lambda a_1, \dots, P_\lambda a_k)$. For fixed $a_j \in E_x$, direct differentiation gives

$$\begin{aligned}\partial_\lambda \widetilde{\mathscr{T}}(a_1, \dots, a_k) = {} & P_\lambda^{-1}(\nabla_{Y_\lambda}\mathscr{T})(P_\lambda a_1, \dots, P_\lambda a_k) \\ & - \mathscr{A}_\lambda \widetilde{\mathscr{T}}(a_1, \dots, a_k) \\ & + \sum_{j=1}^k \widetilde{\mathscr{T}}(a_1, \dots, \mathscr{A}_\lambda a_j, \dots, a_k).\end{aligned}$$

In particular,

$$|\partial_\lambda \widetilde{R}_\lambda| \le |\nabla R|\,|Y_\lambda| + 4|R|\,|\mathscr{A}_\lambda|,$$
$$|\partial_\lambda \widetilde{\nabla R}_\lambda| \le |\nabla^2 R|\,|Y_\lambda| + 5|\nabla R|\,|\mathscr{A}_\lambda|.$$

Substitution of the preceding bounds and integration over $0 \le \lambda \le 1$ yield

$$\sup_s |\widetilde{R}_U(s) - \widetilde{R}_V(s)| \le Cd_0, \qquad \sup_s |\widetilde{\nabla R}_U(s) - \widetilde{\nabla R}_V(s)| \le Cd_0. \tag{2.26}$$

The second inequality uses $|\nabla^{N,2} R^N| \le 1$ for the derivative of the tensor itself. The terms from differentiating parallel transport use only $|R^N|$ and $|\nabla^N R^N|$. This is the only additional curvature derivative required for the two-section estimate. The comparison

uses a family of radial geodesics, and does not require that an exponential map be globally injective.

Step 2. First and vertical Jacobi fields. Define the endomorphism

$$\mathscr{R}_U(s)\xi = \widetilde{R}_U(s)(U,\xi)U.$$

Splitting its difference into the curvature difference and the two differences of U gives

$$|\mathscr{R}_U| \le Cr_U^2, \qquad |\mathscr{R}_U - \mathscr{R}_V| \le Cr_\Sigma d_0. \tag{2.27}$$

For example, the curvature-difference term is bounded by $Cr_U^2 d_0$, and the remaining two terms by $C(r_U + r_V)d_0$; the first is absorbed by the latter after decreasing ε_*. Let E_i^U and K_i^U be the transported fields in (2.22). Their exact integral equations are

$$E_i^U(s) = e_i + sZ_i + \int_0^s (s-\sigma)\mathscr{R}_U(\sigma)E_i^U(\sigma)\,d\sigma,$$
$$K_i^U(s) = Z_i + \int_0^s \mathscr{R}_U(\sigma)E_i^U(\sigma)\,d\sigma.$$

Consequently,

$$|E_i^U| \le C, \quad |K_i^U| \le Cz_U, \quad |E_i^U - e_i - sZ_i| + |K_i^U - Z_i| \le Cr_U^2. \tag{2.28}$$

Subtract the two integral equations, using

$$\mathscr{R}_U E_i^U - \mathscr{R}_V E_i^V = \mathscr{R}_U(E_i^U - E_i^V) + (\mathscr{R}_U - \mathscr{R}_V)E_i^V.$$

To make the absorption explicit, set $E_\Delta = \sup_s |E_i^U(s) - E_i^V(s)|$. The subtracted first integral equation gives

$$E_\Delta \le d_1 + Cr_U^2 E_\Delta + Cr_\Sigma d_0.$$

Choose ε_* so that $Cr_U^2 \le 1/2$. Moving this term to the left yields $E_\Delta \le 2d_1 + Cr_\Sigma d_0 \le Cd$. The second integral equation then gives $\sup_s |K_i^U - K_i^V| \le d_1 + Cr_U^2 E_\Delta + Cr_\Sigma d_0 \le Cd$. Thus

$$\sup_s(|E_i^U - E_i^V| + |K_i^U - K_i^V|) \le Cd, \tag{2.29}$$

$$\sup_s\big(|E_i^U - E_i^V - s(Z_i - W_i)| + |K_i^U - K_i^V - (Z_i - W_i)|\big) \le Cz_\Sigma d. \tag{2.30}$$

Indeed, after subtracting the initial affine terms, the integral is bounded by $C(r_U^2 d + r_\Sigma d_0)$, which is at most $Cz_\Sigma d$.

Let J_U solve

$$J_U'' = \mathscr{R}_U J_U, \qquad J_U(0) = 0, \qquad J_U'(0) = \mathrm{Id}\,.$$

Then $\mathcal{V}_U = J_U(1)$ and

$$J_U(s) = s\,\mathrm{Id} + \int_0^s (s-\sigma)\mathscr{R}_U(\sigma)J_U(\sigma)\,d\sigma.$$

If $J_\Delta = \sup_s |J_U(s) - J_V(s)|$, subtraction of these integral equations, using the uniform bound for J_V, gives

$$J_\Delta \le Cr_U^2 J_\Delta + Cr_\Sigma d_0.$$

Absorbing the first term gives $J_\Delta \le Cr_\Sigma d_0$. The single integral equation also gives $\sup_s |J_U(s) - s\,\mathrm{Id}\,| \le Cr_U^2$. In particular,

$$|\mathcal{V}_U - \mathrm{Id}\,| \le Cr_U^2, \qquad |\mathcal{V}_U - \mathcal{V}_V| \le Cr_\Sigma d_0.$$

The inverse matrices are uniformly bounded. Set $\mathscr{L}_U = \mathcal{V}_U^{-1} - \mathrm{Id}$. The inverse-difference identity

$$\mathcal{V}_U^{-1} - \mathcal{V}_V^{-1} = \mathcal{V}_U^{-1}(\mathcal{V}_V - \mathcal{V}_U)\mathcal{V}_V^{-1}$$

therefore proves

$$|\mathscr{L}_U| \le Cr_U^2, \qquad |\mathscr{L}_U - \mathscr{L}_V| \le Cr_\Sigma d_0. \tag{2.31}$$

Step 3. The induced metric and its exact quadratic remainder. Put $P_i^U = E_i^U(1) - e_i - Z_i$ and $a_{U,ij} = \langle Z_i, e_j\rangle + \langle e_i, Z_j\rangle$. The metric matrix is exactly

$$g_U = I + a_U + R_U^g,$$

where

$$(R_U^g)_{ij} = \langle Z_i, Z_j\rangle + \langle P_i^U, e_j + Z_j\rangle + \langle e_i + Z_i, P_j^U\rangle + \langle P_i^U, P_j^U\rangle.$$

Equations (2.28) and (2.30) imply

$$|R_U^g| \le Cz_U^2, \qquad |R_U^g - R_V^g| \le Cz_\Sigma d, \qquad |a_U - a_V| \le Cd_1.$$

Here and below each product difference is split by replacing one factor at a time. For instance, $\langle Z_i, Z_j\rangle - \langle W_i, W_j\rangle = \langle Z_i - W_i, Z_j\rangle + \langle W_i, Z_j - W_j\rangle$; terms containing $P^U - P^V$ are bounded by (2.30).

Write $X_U = a_U + R_U^g$ and $G_U = g_U^{-1}$. The exact inverse identity

$$G_U = I - X_U + X_U^2 G_U = I - a_U + R_U^G, \qquad R_U^G = -R_U^g + X_U^2 G_U,$$

together with $G_U - G_V = G_U(X_V - X_U)G_V$, gives

$$|G_U - I| \le Cz_U, \quad |G_U - G_V| \le Cd, \quad |R_U^G| \le Cz_U^2, \quad |R_U^G - R_V^G| \le Cz_\Sigma d.$$

For the last estimate one uses explicitly

$$X_U^2 G_U - X_V^2 G_V = \big[(X_U - X_V)X_U + X_V(X_U - X_V)\big]G_U + X_V^2(G_U - G_V).$$

No Taylor polynomial replaces the inverse matrix in these identities.

Define endomorphism-valued coefficients

$$\mathcal{M}_U^{ij} = G_U^{ij}\mathcal{V}_U^{-1}, \qquad \mathcal{D}_U^{ij} = \mathcal{M}_U^{ij} - \delta^{ij}\,\mathrm{Id} + a_U^{ij}\,\mathrm{Id}\,.$$

The identity $\mathcal{D}_U^{ij} = (R_U^G)^{ij}\,\mathrm{Id} + G_U^{ij}\mathscr{L}_U$ and (2.31) give

$$|\mathcal{M}_U - \delta\,\mathrm{Id}\,| \le Cz_U, \qquad |\mathcal{D}_U| \le Cz_U^2, \tag{2.32}$$

$$|G_U - G_V| + |\mathcal{M}_U - \mathcal{M}_V| \le Cd, \tag{2.33}$$

$$|\mathcal{D}_U - \mathcal{D}_V| \le Cz_\Sigma d. \tag{2.34}$$

For example, $G_U\mathscr{L}_U - G_V\mathscr{L}_V = (G_U - G_V)\mathscr{L}_U + G_V(\mathscr{L}_U - \mathscr{L}_V)$; both terms have the stated quadratic-difference bound.

Step 4. The exact second-variation remainder. Set $B_{ij}^U = R_q^N(U, e_i)e_j$. This is linear in U, so

$$|B^U| \le Cr_U, \qquad |B^U - B^V| \le Cd_0.$$

In the fixed fiber, the inhomogeneous term in (2.22) is

$$\begin{aligned}\mathcal{F}_{ij}^U = {} & \widetilde{\nabla R}_U(E_i^U; U, E_j^U, U) + \widetilde{\nabla R}_U(U; U, E_i^U, E_j^U) \\ & + \widetilde{R}_U(K_i^U, E_j^U)U + \widetilde{R}_U(U, E_j^U)K_i^U + 2\widetilde{R}_U(U, E_i^U)K_j^U + \widetilde{R}_U(U, K_i^U)E_j^U,\end{aligned}$$

where $(\nabla R)(X; Y, Z, W) = (\nabla_X R)(Y, Z)W$. Each of the first two terms has two factors of U; each of the last four has one factor of U and one factor of K. Therefore $|\mathcal{F}^U| \le Cz_U^2$.

We record the difference estimates for these products. For a term containing $\widetilde{\nabla R}$, replace first the tensor, then the two U factors, and finally the two E factors. Equations (2.26) and (2.29) give, for either such term,

$$|\text{difference}| \le C\big((r_U^2 + r_V^2)d_0 + r_\Sigma d_0 + (r_U^2 + r_V^2)d\big) \le C z_\Sigma d.$$

For example, a curvature term $\widetilde{R}(K_i, E_j)U$ has difference bounded by

$$C\big(r_U z_U d_0 + r_U d + r_U z_V d + z_V d_0\big) \le C z_\Sigma d,$$

by replacing, in order, $\widetilde{R}, K_i, E_j, U$. The other three curvature terms obey the same estimate, with the indicated factors permuted. It follows that

$$\sup_s |\mathcal{F}^U(s) - \mathcal{F}^V(s)| \le C z_\Sigma d. \tag{2.35}$$

Let $H^U_{ij} = (\nabla^{E,2}U)_{ij}$ and define

$$S^U_{ij}(s) = C^U_{ij}(s) - J_U(s)H^U_{ij} - A_{ij} - sB^U_{ij}.$$

The part $J_U H^U_{ij}$ solves the homogeneous Jacobi equation. Thus $S^U(0) = (S^U)'(0) = 0$ and the exact integral equation is

$$S^U_{ij}(s) = \int_0^s (s-\sigma)\left\{\mathscr{R}_U(\sigma)\big(A_{ij} + \sigma B^U_{ij} + S^U_{ij}(\sigma)\big) + \mathcal{F}^U_{ij}(\sigma)\right\} d\sigma. \tag{2.36}$$

In particular, S^U is independent of H^U. Let $s_U = \sup_s |S^U(s)|$. Since $|A_0| \le 1$, the integral equation gives

$$s_U \le C r_U^2(1 + r_U + s_U) + C z_U^2 \le C r_U^2 s_U + C z_U^2.$$

With $C r_U^2 \le 1/2$, absorption yields $s_U \le C z_U^2$. The difference of the integrands in (2.36) is exactly

$$\begin{aligned}\mathscr{R}_U(S^U - S^V) + (\mathscr{R}_U - \mathscr{R}_V)(A + \sigma B^V + S^V)\\ + \sigma\mathscr{R}_U(B^U - B^V) + \mathcal{F}^U - \mathcal{F}^V.\end{aligned}$$

The second term has norm at most $C r_\Sigma d_0(1 + r_V + z_V^2)$, the third at most $C r_U^2 d_0$, and the fourth at most $C z_\Sigma d$. Thus, with $s_\Delta = \sup_s |S^U(s) - S^V(s)|$, integration gives

$$s_\Delta \le C r_U^2 s_\Delta + C z_\Sigma d.$$

Absorbing $C r_U^2 s_\Delta$ gives

$$\sup_{0\le s\le 1} |S^U(s) - S^V(s)| \le C z_\Sigma d. \tag{2.37}$$

Step 5. The full remainder and its difference. At $s = 1$, the identity

$$C^U_{ij}(1) = \mathcal{V}_U H^U_{ij} + A_{ij} + B^U_{ij} + S^U_{ij}(1)$$

gives the exact formula

$$\begin{aligned}\mathcal{Q}(U) = (G_U^{ij} - \delta^{ij})H^U_{ij} + \mathcal{D}^{ij}_U A_{ij}\\ + (\mathcal{M}^{ij}_U - \delta^{ij}\,\mathrm{Id})B^U_{ij} + \mathcal{M}^{ij}_U S^U_{ij}(1).\end{aligned} \tag{2.38}$$

Indeed, the subtracted constant is $\delta^{ij}A_{ij} = \mathbf{H}_0$; the subtracted linear terms are $\delta^{ij}H^U_{ij} + \delta^{ij}B^U_{ij} - a_U^{ij}A_{ij} = L_0 U$. Thus (2.38) is an identity for the full operator, not for a finite Taylor expansion. It also verifies the one-section bound (2.21): the four terms are bounded, respectively, by $C z_U h_U$, $C z_U^2$, $C z_U r_U$, and $C z_U^2$. These individual size bounds use only R^N

and $\nabla^N R^N$; the second curvature derivative entered only in comparing the two transported tensors in Step 1. In particular the remainder and its first variation vanish at the zero jet. More explicitly, scaling the entire jet (U, Z, H) by λ gives a remainder of size $C\lambda^2(z_U^2 + z_U h_U)$.

We take the difference of the four terms separately. The first equals

$$(G_U - I)(H^U - H^V) + (G_U - G_V)H^V,$$

and its norm is at most $Cz_U d_2 + Ch_V d$. The second has norm at most $Cz_\Sigma d$, by (2.34) and $|A_0| \le 1$. For the third, use the exact splitting

$$(\mathcal{M}_U - I)B^U - (\mathcal{M}_V - I)B^V = (\mathcal{M}_U - \mathcal{M}_V)B^U + (\mathcal{M}_V - I)(B^U - B^V).$$

Its norm is bounded by $Cr_U d + Cz_V d_0 \le Cz_\Sigma d$. Finally,

$$\mathcal{M}_U S^U - \mathcal{M}_V S^V = (\mathcal{M}_U - \mathcal{M}_V)S^U + \mathcal{M}_V(S^U - S^V),$$

whose norm is at most $Cz_U^2 d + Cz_\Sigma d$ by (2.37). Adding these bounds and, if necessary, increasing the right-hand side symmetrically in U, V proves (2.24).

Step 6. The weighted estimate. Let $D = \|U - V\|_{\mathcal{X}_T}$. The definitions give

$$\|d\|_\infty \le CD, \quad \|d\|_{\mathcal{Y}_T} \le C\sqrt{T}\, D, \quad \|d_2\|_{\mathcal{Y}_T} \le D, \quad \|z_\Sigma\|_\infty \le 2\delta, \quad \|h_U + h_V\|_{\mathcal{Y}_T} \le 2\delta.$$

Therefore $\|\mathcal{Q}(U) - \mathcal{Q}(V)\|_{\mathcal{Y}_T} \le C\delta D$. The operator

$$\mathcal{D}_0 w = \sum_i R^N(w, e_i)e_i - 2\sum_{i,j}\langle \nabla^E_{X_i} w, e_j\rangle A_{ij}$$

has bounded coefficients and order at most one, so $\|\mathcal{D}_0(U-V)\|_{\mathcal{Y}_T} \le C\sqrt{T}\, D$. This proves (2.25). All the pointwise estimates are inequalities for the finite-dimensional jets. They therefore hold almost everywhere for sections in $\mathcal{X}_T$ with weak Hessians, and give the same weighted estimate in that space. □

2.3. **Evolution of the second fundamental form.** We recall the standard evolution estimate for the second fundamental form. For a mean curvature flow in a fixed ambient manifold, the tensor evolution equation has the following form

$$(\partial_t - \Delta)A = A * A * A + \mathrm{Rm}_N * A + \nabla^N \mathrm{Rm}_N\,. \tag{2.39}$$

Combining Kato's inequality and Young's inequality, we get

$$\begin{aligned}(\partial_t - \Delta)|A|^2 &\le -2|\nabla A|^2 + C\big(|A|^4 + K_0|A|^2 + K_1|A|\big)\\ &\le -|\nabla A|^2 + C\big(|A|^4 + K_0^2 + K_1^{4/3}\big).\end{aligned} \tag{2.40}$$

The curvature bound for the solution constructed below is proved directly by uniform local parabolic estimates in Lemma 6.3. This provides a bounded-curvature interval starting at the initial time. The evolution inequality (2.40) will also be used in the derivative estimates for the uniqueness argument.

3. Coulomb gauges and harmonic coordinates

3.1. **Coulomb gauges.** We derive the Coulomb gauge equation from changes of orthonormal frame and from a variational principle. We then construct a quantitative local gauge for the pullback tangent bundle of an immersed submanifold. We recall the Euclidean Uhlenbeck theorem [12] and give the controlled-metric version needed below, including dimensions one and two.

Let (E, h, ∇^E) be a rank-m Euclidean vector bundle with a metric connection over a Riemannian manifold (M^n, g). Choose a coordinate ball B on which E is trivial, and choose an orthonormal frame $e = (e_1, \dots, e_m)$. We identify a section $s = e_\alpha u^\alpha$ with the vector $u = (u^\alpha)$. We use the convention

$$\nabla^E_i s = e\big(\partial_i u + \omega_i u\big), \qquad \omega = \omega_i \, \mathrm{d}x^i \in \Omega^1(B; \mathfrak{so}(m)). \tag{3.1}$$

If $Q : B \to SO(m)$ and $e' = eQ$, the coordinate vectors satisfy $u = Qu'$. Differentiating this identity in (3.1) gives

$$\omega' = Q^{-1}\omega Q + Q^{-1}\,\mathrm{d}Q. \tag{3.2}$$

The curvature form is

$$\Omega = \mathrm{d}\omega + \omega \wedge \omega, \qquad \Omega' = Q^{-1}\Omega Q. \tag{3.3}$$

Here $(\omega\wedge\omega)_{ij} = [\omega_i, \omega_j]$. In particular, $|\Omega|$ is independent of the chosen orthonormal frame.

A *Coulomb frame* on B is a frame in which

$$\mathrm{d}^*_g \omega' = 0. \tag{3.4}$$

Our convention is $\mathrm{d}^*_g \omega' = -\nabla^i \omega'_i$, with the Levi–Civita covariant derivative acting on the one-form index. A commonly used accompanying boundary condition is

$$\iota_\nu \omega' = 0 \quad \text{on } \partial B, \tag{3.5}$$

where ν denotes the outward unit normal to the boundary. Thus gauge fixing amounts to solving the nonlinear system

$$\begin{cases} \mathrm{d}^*_g\big(Q^{-1}\omega Q + Q^{-1}\,\mathrm{d}Q\big) = 0 & \text{in } B, \\ \iota_\nu\big(Q^{-1}\omega Q + Q^{-1}\,\mathrm{d}Q\big) = 0 & \text{on } \partial B. \end{cases} \tag{3.6}$$

The second line can be omitted if another boundary normalization is selected; it is particularly convenient for the div–curl estimates below.

Consider the functional

$$\mathcal{E}(Q) = \frac{1}{2}\int_B |\omega^Q|^2 \, \mathrm{d}\mu_g, \qquad \omega^Q = Q^{-1}\omega Q + Q^{-1}\,\mathrm{d}Q. \tag{3.7}$$

Let $Q_s = Q\exp(s\xi)$ for $\xi : B \to \mathfrak{so}(m)$, and set $a = \omega^Q$. Direct differentiation yields

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \omega^{Q_s} = \mathrm{d}\xi + [a, \xi]. \tag{3.8}$$

For the invariant inner product on $\mathfrak{so}(m)$, $\langle a_i, [a_i, \xi]\rangle = 0$ pointwise. Integration by parts therefore gives

$$\begin{aligned} \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \mathcal{E}(Q_s) &= \int_B \langle a, \, \mathrm{d}\xi\rangle \, \mathrm{d}\mu_g \\ &= \int_B \langle \, \mathrm{d}^*_g a, \xi\rangle \, \mathrm{d}\mu_g + \int_{\partial B} \langle \iota_\nu a, \xi\rangle \, \mathrm{d}\sigma_g. \end{aligned} \tag{3.9}$$

Consequently, a smooth stationary point with variations free on the boundary satisfies (3.4)–(3.5). This calculation explains the condition; it does not by itself establish that a minimizing gauge has the regularity or quantitative bounds needed for analysis.

There is also a useful linearization. Formally set $Q = \exp \xi$ near the identity. Then

$$\omega^Q = \omega + \mathrm{d}\xi + [\omega, \xi] + O\big(|\xi|\,|\mathrm{d}\xi| + |\xi|^2|\omega|\big), \tag{3.10}$$

and the principal part of (3.6) is

$$\mathrm{d}_g^* \mathrm{d}\xi = -\mathrm{d}_g^* \omega, \qquad \partial_\nu \xi = -\iota_\nu \omega. \tag{3.11}$$

This is a Neumann problem for the scalar Laplacian acting componentwise on $\mathfrak{so}(m)$; constant gauge transformations constitute its finite-dimensional kernel. A normalization, for example zero average for ξ, removes this ambiguity in a local implicit-function argument.

Here is a convenient form of the local gauge-fixing result.

Theorem 3.1 (Uhlenbeck [12])**.** *Let $n \geq 3$, $p > n$, and let $E \to B_r \subset \mathbb{R}^n$ be a rank-m Euclidean vector bundle with a smooth metric connection ∇^E. Denote its curvature by Ω, and suppose*

$$\|\Omega\|_{L^\infty(B_r)} \leq K, \qquad r^2 K \leq \varepsilon(n, m, p).$$

Then E admits an orthonormal frame on B_r in which the connection form ω satisfies

$$d^*\omega = 0 \quad \text{in } B_r, \qquad \iota_\nu \omega = 0 \quad \text{on } \partial B_r,$$

and

$$r^{-1-n/p}\|\omega\|_{L^p(B_r)} + r^{-n/p}\|\nabla\omega\|_{L^p(B_r)} \leq C(n, m, p)K.$$

Lemma 3.2 (Coulomb frames for controlled metrics)**.** *Let $n \geq 1$, let $p > n$, and let g be a smooth metric on $B_2 \subset \mathbb{R}^n$ satisfying*

$$\tfrac{1}{2}\delta \leq g \leq 2\delta, \qquad \|\partial g\|_{L^p(B_2)} + \|\partial^2 g\|_{L^p(B_2)} \leq 1.$$

Let $E \to B_2$ be a smooth rank-r Euclidean bundle with a smooth metric connection and curvature R^E. There are constants $\varepsilon = \varepsilon(n, r, p) > 0$ and $C = C(n, r, p)$ such that, if

$$\kappa := \|R^E\|_{L^\infty(B_2, g)} \leq \varepsilon,$$

then E has a smooth orthonormal frame on $B_{3/2}$ whose connection form a satisfies

$$d_g^* a = 0, \qquad \|a\|_{W^{1,p}(B_{3/2})} + \|a\|_{L^\infty(B_{3/2})} \leq C\kappa. \tag{3.12}$$

No boundary condition for a is asserted. The lemma holds in dimensions one and two as well.

Proof. Choose an orthonormal frame at the origin and extend it by parallel transport along the coordinate radial segments. In this radial frame, write the connection and curvature forms as b and F_b. The radial gauge identity is

$$x^i b_i(x) = 0, \qquad b_i(x) = \int_0^1 t x^j (F_b)_{ji}(tx)\, dt.$$

Metric equivalence and orthogonality of parallel transport imply

$$\|b\|_{L^\infty(B_2)} \leq C\kappa. \tag{3.13}$$

This estimate uses no derivative of the curvature.

Choose a finite exponent $q > \max\{2n, 2p, 4\}$. The metric assumptions give a uniform $C^{1,\alpha}$ bound, where $\alpha = 1 - n/p > 0$. The Dirichlet operator $d_g^* d$, acting componentwise on $\mathfrak{so}(r)$-valued functions, therefore has the estimate

$$\|\xi\|_{W^{1,q}(B_2)} \le C_q \|v\|_{L^q(B_2)}, \qquad d_g^* d\xi = -d_g^* v, \quad \xi|_{\partial B_2} = 0. \tag{3.14}$$

This is the usual divergence-form Dirichlet estimate on a fixed Euclidean ball with uniformly elliptic Lipschitz coefficients; its constants are controlled by the displayed metric bounds and q. Write the solution operator in (3.14) as $\mathcal{T}_g v$.

For $\xi \in W_0^{1,q}(B_2; \mathfrak{so}(r))$, set $Q = \exp \xi$ and expand

$$Q^{-1} b Q + Q^{-1} dQ = b + d\xi + \mathcal{R}(\xi, b).$$

The matrix exponential and the embedding $W^{1,q} \hookrightarrow L^\infty$ give, on a sufficiently small $W^{1,q}$ ball,

$$\begin{aligned} \|\mathcal{R}(\xi, b)\|_{L^q} &\le C\big(\|\xi\|_{W^{1,q}}^2 + \|b\|_{L^q} \|\xi\|_{W^{1,q}}\big), \\ \|\mathcal{R}(\xi, b) - \mathcal{R}(\eta, b)\|_{L^q} &\le C\big(\|\xi\|_{W^{1,q}} + \|\eta\|_{W^{1,q}} + \|b\|_{L^q}\big) \|\xi - \eta\|_{W^{1,q}}. \end{aligned}$$

Consequently,

$$\xi \longmapsto \mathcal{T}_g\big(b + \mathcal{R}(\xi, b)\big)$$

is a contraction on a ball of radius $C\kappa$ in $W_0^{1,q}$, provided κ is sufficiently small. Its fixed point defines a gauge in which

$$a = Q^{-1} b Q + Q^{-1} dQ, \qquad d_g^* a = 0, \qquad \|a\|_{L^q(B_2)} \le C\kappa.$$

The curvature identity holds distributionally:

$$da = Q^{-1} F_b Q - a \wedge a.$$

We use the interior div–curl estimate

$$\|v\|_{W^{1,s}(B_\rho)} \le C\big(\|dv\|_{L^s(B_R)} + \|d_g^* v\|_{L^s(B_R)} + \|v\|_{L^s(B_R)}\big), \qquad 1 < s < \infty, \quad \rho < R < 2.$$

Its constant is uniform for the present metrics. Indeed, freeze the principal coefficients on sufficiently small coordinate balls, apply the constant-coefficient div–curl estimate, and absorb the coefficient oscillation. The terms containing ∂g are zeroth-order terms. A finite covering of the fixed interior ball completes the estimate. Applying it first with $s = q/2$ gives

$$\|a\|_{W^{1,q/2}(B_{7/4})} \le C(\kappa + \kappa^2) \le C\kappa.$$

Since $q/2 > n$, it follows that $\|a\|_{L^\infty(B_{7/4})} \le C\kappa$. A second application, now with $s = p$, gives

$$\|a\|_{W^{1,p}(B_{3/2})} \le C\big(\kappa + \|a\|_{L^\infty(B_{7/4})}^2 + \|a\|_{L^p(B_{7/4})}\big) \le C\kappa.$$

These estimates also justify the weak div–curl applications by approximation. Finally, $dQ = Qa - bQ$, the curvature identity, and interior elliptic regularity bootstrap Q and a to smoothness on $B_{3/2}$. Only their smoothness, and not uniform higher derivative estimates, is used in this final bootstrap. □

3.2. **Harmonic coordinates.** We next recall the harmonic radius used in the local estimates.

Definition 3.3 ($W^{k,p}$ harmonic radius)**.** Suppose (M^n, g) is a Riemannian manifold, $n < p < \infty$, and $x \in M$. The W^k_p-harmonic radius at x is the supremum of all $R > 0$ such that there exists a coordinate chart $\phi : B_R(x) \to \mathbb{R}^n$ satisfying

- $\Delta\phi^j = 0$ on $B_R(x)$ and $\phi^j(x) = 0$ for each j;
- $\frac{1}{2}\delta_{ij} \le g_{ij} \le 2\delta_{ij}$, in $B_R(x)$ as symmetric bilinear forms;
- $\sum_{1\le|J|\le k} R^{|J|-\frac{n}{p}} \|\partial^J g_{ij}\|_{L^p(B_R(x))} \le 1$.

We denote the W^k_p-harmonic radius at x by $r_{k,p}(x)$.

Lemma 3.4. *Fix $p > n$. There is a constant $\xi = \xi(n,p) \in (0, \frac{1}{100n\pi})$ with the following property. Suppose (M, g) is a Riemannian manifold satisfying $|Rm| \le \xi^2$. Suppose $x \in M$ and $B_{100n\pi}(0) \subset T_xM$ is equipped with the pull-back metric $\tilde{g} = Exp_x^* g$. Then the $W^{2,p}$-harmonic radius of 0 is at least 100.*

Proof. The curvature bound and the choice of ξ ensure that the exponential map has no conjugate points on the indicated tangent ball. Its pullback metric is therefore smooth. By the Gauss lemma, geodesics issuing from 0 are radial and the injectivity radius at 0, relative to this ball, is bounded below by a fixed dimensional multiple of its radius. The usual harmonic-radius estimate under a sectional-curvature bound and a positive injectivity-radius bound gives a $W^{2,p}$ harmonic chart at 0. After scaling and decreasing $\xi(n,p)$, its radius is at least 100. This argument takes place on the tangent ball and does not require a lower bound for the injectivity radius of M. □

3.3. **The rough Laplacian in harmonic coordinates and a Coulomb frame.** For a section $U = e_\alpha u^\alpha$,

$$\nabla^E_i U = e_\alpha \big(\partial_i u + \omega_i u\big)^\alpha.$$

Recall the rough Laplacian is defined by

$$\Delta^E U = g_0^{ij}\left(\nabla^E_i \nabla^E_j U - \Gamma(g_0)^k_{ij} \nabla^E_k U\right).$$

Lemma 3.5. *Let $F_0 : (M^n, g_0) \to (\bar{M}^{n+m}, \bar{g})$ be a smooth isometric immersion, and equip*

$$E := F_0^* T\bar{M}$$

with the pullback metric and connection ∇^E. Fix a finite exponent $p > n$.

Suppose that a coordinate neighborhood is identified with $B_{2r} \subset \mathbb{R}^n$ through g_0-harmonic coordinates. Assume that the rescaled metric

$$\widetilde{g}_{ij}(y) := (g_0)_{ij}(ry)$$

satisfies

$$2^{-1}\delta \le \widetilde{g} \le 2\delta, \qquad \sum_{1\le|J|\le 2} \|\partial^J \widetilde{g}\|_{L^p(B_2)} \le 1. \tag{3.15}$$

Assume that

$$|\overline{\mathrm{Rm}}| \circ F_0 \le K_0, \qquad |\bar{\nabla}\overline{\mathrm{Rm}}| \circ F_0 \le K_1, \qquad |A_0| \le \Lambda \quad \text{on } B_{2r}, \tag{3.16}$$

where A_0 is the second fundamental form of F_0.

There exist constants $\varepsilon_p > 0$ and $C_p < \infty$, depending only on n, m, p, such that, if $r^2 K_0 \le \varepsilon_p$, then $E|_{B_r}$ admits an orthonormal frame $e_1, \dots, e_{n+m}$ whose connection form

$$\nabla^E_i e_\beta = e_\alpha (\omega_i)^\alpha{}_\beta$$

satisfies the Coulomb condition

$$\mathrm{d}^*_{g_0} \omega = 0 \quad \text{on } B_r.$$

In this frame,

$$\sum_{j=0}^{2} r^{j+1-n/p} \|\partial^j \omega\|_{L^p(B_r(x))} \le C_p \left[r^2 K_0 + r^3 (K_1 + \Lambda K_0) \right]. \tag{3.17}$$

The same estimates hold with ∂ replaced by the Levi–Civita covariant derivative on the one-form index and with norms taken using g_0.

For a section $U = e_\alpha u^\alpha$, the rough Laplacian has the following expression on B_r:

$$(\Delta^E U)^\alpha = g_0^{ij} \partial_i \partial_j u^\alpha + 2 g_0^{ij} (\omega_i)^\alpha{}_\beta \partial_j u^\beta + g_0^{ij} (\omega_i)^\alpha{}_\gamma (\omega_j)^\gamma{}_\beta u^\beta. \tag{3.18}$$

Proof. We use the curvature convention

$$\bar{R}(X, Y) = \bar{\nabla}_X \bar{\nabla}_Y - \bar{\nabla}_Y \bar{\nabla}_X - \bar{\nabla}_{[X,Y]}.$$

The curvature of the pullback connection is

$$R^E(X, Y) = \bar{R}(\,\mathrm{d}F_0(X),\, \mathrm{d}F_0(Y)).$$

Since F_0 is isometric,

$$|R^E| \le C(n, m) K_0. \tag{3.19}$$

Let $\mathcal{D} R^E$ denote the covariant derivative of R^E. The connections used are the induced connection on $\mathrm{End}(E)$ and the Levi–Civita connection on its two covariant indices. Using the Gauss formula

$$\bar{\nabla}_Z \,\mathrm{d}F_0(X) = \,\mathrm{d}F_0(\nabla^{g_0}_Z X) + A_0(Z, X),$$

we obtain

$$\begin{aligned} (\mathcal{D}_Z R^E)(X, Y) = {} & (\bar{\nabla}_{\mathrm{d}F_0(Z)} \bar{R})(\,\mathrm{d}F_0(X),\, \mathrm{d}F_0(Y)) \\ & + \bar{R}(A_0(Z, X),\, \mathrm{d}F_0(Y)) \\ & + \bar{R}(\,\mathrm{d}F_0(X), A_0(Z, Y)). \end{aligned} \tag{3.20}$$

Hence,

$$|\mathcal{D} R^E| \le C(n, m)(K_1 + \Lambda K_0). \tag{3.21}$$

Rescale the coordinates by $x = ry$, and set

$$\kappa := r^2 K_0, \qquad \delta := r^3 (K_1 + \Lambda K_0).$$

If Ω is the curvature form, the rescaled connection and curvature forms are

$$a_i(y) = r \omega_i(ry), \qquad \Phi_{ij}(y) = r^2 \Omega_{ij}(ry).$$

Equations (3.19) and (3.21) give

$$\|\Phi\|_{L^p(B_2)} \le C\kappa, \qquad \|\mathcal{D}\Phi\|_{L^p(B_2)} \le C\delta.$$

All coordinate and metric norms on the rescaled ball are uniformly equivalent.

For sufficiently small κ, Lemma 3.2 applied to $\widetilde{g}$ and the rescaled pullback connection gives an orthonormal frame on $B_{3/2}$ such that

$$d^*_{\widetilde{g}} a = 0, \qquad \|a\|_{W^{1,p}(B_{3/2})} + \|a\|_{L^\infty(B_{3/2})} \le C_p \kappa. \tag{3.22}$$

The argument is valid for every $n \ge 1$. The subsequent estimates are interior estimates, so no boundary normalization of this frame is needed.

We now estimate the second derivatives. In the chosen frame, the full covariant derivative of the curvature form is

$$(\mathcal{D}_k \Phi)_{ij} = \partial_k \Phi_{ij} - \widetilde{\Gamma}^\ell_{ki} \Phi_{\ell j} - \widetilde{\Gamma}^\ell_{kj} \Phi_{i\ell} + [a_k, \Phi_{ij}].$$

In the $W^{2,p}$ harmonic coordinates, we have

$$\|\widetilde{\Gamma}\|_{L^\infty(B_{3/2})} + \|\partial \widetilde{\Gamma}\|_{L^p(B_{3/2})} \le C.$$

It follows that

$$\begin{aligned} \|\partial \Phi\|_{L^p(B_{3/2})} &\le C \left(\|\mathcal{D}\Phi\|_{L^p(B_{3/2})} + \|\widetilde{\Gamma}\|_{L^\infty(B_{3/2})} \|\Phi\|_{L^p(B_{3/2})} + \|a\|_{L^\infty(B_{3/2})} \|\Phi\|_{L^p(B_{3/2})} \right) \\ &\le C_p(\delta + \kappa + \kappa^2). \end{aligned}$$

Thus

$$\|\Phi\|_{W^{1,p}(B_{3/2})} \le C_p(\delta + \kappa + \kappa^2). \tag{3.23}$$

The connection form satisfies the first-order elliptic system

$$\mathrm{d} a = \Phi - a \wedge a, \qquad \mathrm{d}^*_{\widetilde{g}} a = 0.$$

The interior estimate for this system gives

$$\|a\|_{W^{2,p}(B_1)} \le C_p \big(\|\,\mathrm{d} a\|_{W^{1,p}(B_{3/2})} + \|\,\mathrm{d}^*_{\widetilde{g}} a\|_{W^{1,p}(B_{3/2})} + \|a\|_{W^{1,p}(B_{3/2})} \big).$$

Here the coefficient bounds required for the estimate follow from (3.15). Indeed, the principal coefficients are uniformly elliptic and Lipschitz, and their second derivatives belong to L^p; the resulting products are controlled using $W^{1,p} \hookrightarrow L^\infty$. More explicitly, apply the first-order div–curl estimate to each $\partial_k a$. Commuting ∂_k through $d^*_{\widetilde{g}}$ produces terms of the forms $(\partial \widetilde{g}) * \partial a$ and $(\partial^2 \widetilde{g}) * a$, with bounded factors of $\widetilde{g}^{-1}$ understood. They belong to L^p because $\partial \widetilde{g}, a \in L^\infty$ and $\partial a, \partial^2 \widetilde{g} \in L^p$. The curl equation contributes $\partial \Phi - \partial(a \wedge a)$. Thus only one covariant derivative of the bundle curvature is needed.

Moreover,

$$\|a \wedge a\|_{W^{1,p}(B_{3/2})} \le C_p \|a\|_{L^\infty(B_{3/2})} \|a\|_{W^{1,p}(B_{3/2})} \le C_p \kappa^2.$$

Combining this with (3.22) and (3.23), and using the smallness of κ, we obtain

$$\|a\|_{W^{2,p}(B_1)} \le C_p(\kappa + \delta). \tag{3.24}$$

By rescaling, we obtain (3.17). For the covariant versions, use

$$\widetilde{\nabla} a = \partial a + \widetilde{\Gamma} * a$$

and

$$\widetilde{\nabla}^2 a = \partial^2 a + \widetilde{\Gamma} * \partial a + (\partial \widetilde{\Gamma}) * a + \widetilde{\Gamma} * \widetilde{\Gamma} * a.$$

The bounds on $\widetilde{\Gamma}$, $\partial \widetilde{\Gamma}$, and a established above control every term in L^p. Scaling back proves the claimed covariant estimates.

In local coordinates, we have

$$\begin{aligned}(\Delta^E U)^\alpha = {} & g_0^{ij}\partial_i\partial_j u^\alpha - g_0^{ij}\Gamma(g_0)_{ij}^k\partial_k u^\alpha + 2g_0^{ij}(\omega_i)^\alpha{}_\beta\partial_j u^\beta \\ & + g_0^{ij}\left(\partial_i(\omega_j)^\alpha{}_\beta - \Gamma(g_0)_{ij}^k(\omega_k)^\alpha{}_\beta + (\omega_i)^\alpha{}_\gamma(\omega_j)^\gamma{}_\beta\right)u^\beta.\end{aligned} \tag{3.25}$$

Here $(\Delta^E U)^\alpha$ denotes the component of $\Delta^E U$ in the frame e_α. On the other hand, in B_r, the harmonic-coordinate condition gives

$$g_0^{ij}\Gamma(g_0)_{ij}^k = 0, \tag{3.26}$$

while the Coulomb condition gives

$$\nabla^i\omega_i = g_0^{ij}\left(\partial_i\omega_j - \Gamma(g_0)_{ij}^k\omega_k\right) = 0. \tag{3.27}$$

Combining (3.25)–(3.27) gives (3.18). In particular, the potentially differentiated zeroth-order coefficient $g_0^{ij}\partial_i\omega_j$ has disappeared by the Coulomb condition. The remaining coefficients of (3.18) have two coordinate derivatives in L^p, with constants depending only on K_0, K_1, Λ and the stated metric bounds. Neither ∇A_0 nor a second derivative of the ambient curvature occurs in this conclusion. □

4. Heat kernel estimates

Throughout this section the linear operator is the fixed rough Laplacian Δ^E on $E = F_0^*TN$. Its heat kernel is denoted by K_E. Although the full linearization is

$$L_0 = \Delta^E + B_0\nabla^E + C_0, \qquad |B_0| + |C_0| \le C(n, m, K_0, \Lambda),$$

B_0 contains A_0. A bound for A_0 provides no uniform bound for its derivatives. Accordingly, the equation solved below is written as

$$(\partial_t - \Delta^E)U = \mathbf{H}_0 + (L_0 - \Delta^E)U + \mathcal{Q}(U).$$

All lower-order terms of $L_0 - \Delta^E$ remain on the right-hand side. We construct no higher-order heat-kernel estimates for L_0 and never differentiate A_0 in the linear theory. The estimates in this section use only K_0, K_1, Λ; the second ambient curvature-derivative bound used in the nonlinear contraction has a separate role.

Definition 4.1. Suppose $0 < \alpha < 1$, $r > 0$, and l is a nonnegative integer. Suppose s is a smooth section of a vector bundle $\mathcal{E} \to M$. Define

$$[s]_{l+\alpha}^{(r)}(x) := \sup_{0<d(x,x')<r}\ \sup_{\gamma\in\Xi(x,x')} \frac{|\nabla^l s(x) - \tau_\gamma(x,x')\nabla^l s(x')|}{d^\alpha(x,x')}, \tag{4.1}$$

where

$$\Xi(x,x') := \{\text{the shortest geodesic segment connecting } x \text{ and } x'\}, \tag{4.2}$$

and τ_γ is parallel transport from x' to x in the tensor bundle $T^{*\otimes l}M \otimes \mathcal{E}$. The seminorm is allowed to be infinite.

The metric comparison in Definition 3.3, applied to radial coordinate segments and minimizing geodesics, gives

$$B_{0.5R}(0) \subset B_{\frac{\sqrt{2}}{2}R}(0) \subset \phi_R(B_R(x)) \subset B_{\sqrt{2}R}(0) \tag{4.3}$$

for every $R < r_{k,p}(x)$. Therefore, the Euclidean ball $B_{0.5R}(0) \subset \mathbb{R}^n$ is equipped with two metrics, the Euclidean metric g_E and the push-forward metric $(\phi_R^{-1})^* g$. For simplicity of notation, we still denote $(\phi_R^{-1})^* g$ by g. In other words, we can regard the identity map from $(B_{0.5R}(0), g)$ to $(B_{0.5R}(0), g_E)$ as a harmonic map. Thus, we have

$$0 = \Delta_g x^k = g^{ij}\left(\frac{\partial^2 x^k}{\partial x^i \partial x^j} - \Gamma_{ij}^l \frac{\partial x^k}{\partial x^l}\right) = -g^{ij}\Gamma_{ij}^k. \tag{4.4}$$

The tensor g_{ij} is now a matrix-valued function. In light of the relationship (4.3), it follows from the definition of the harmonic radius that the following inequalities hold.

$$\frac{1}{2}\sum_{i=1}^n (V^i)^2 \le g_{ij} V^i V^j \le 2\sum_{i=1}^n (V^i)^2, \quad \forall\, V \in \mathbb{R}^n; \tag{4.5}$$

$$\sum_{1\le |J|\le k} R^{|J|-\frac{n}{p}} \|\partial^J g_{ij}\|_{L^p(B_{R/2}(0))} \le 1. \tag{4.6}$$

Suppose $R \ge 100$. Then $B_{0.5R}(0) \supset B_2(0)$. Define

$$\begin{cases} B := B_2(0), \quad B' := B_1(0); \\ \Omega := B \times [-4, 0], \quad \Omega' := B' \times [-1, 0]. \end{cases} \tag{4.7}$$

We use the standard parabolic Sobolev and Hölder norms. In particular,

$$\|u\|_{W_p^{2m,m}(\Omega)} := \sum_{|J|+2k\le 2m} \|D^J \partial_t^k u\|_{L^p(\Omega)}, \tag{4.8}$$

$$\begin{aligned} \|u\|_{C^{m+\alpha,(m+\alpha)/2}(\Omega)} := &\sum_{|J|+2k\le m} \|D^J \partial_t^k u\|_{L^\infty(\Omega)} + \sum_{|J|+2k=m} [D^J \partial_t^k u]_{\alpha,\alpha/2;\Omega} \\ &+ \sum_{|J|+2k=m-1} [D^J \partial_t^k u]_{t;(1+\alpha)/2;\Omega}. \end{aligned} \tag{4.9}$$

Here J is a spatial multi-index, $[\cdot]_{\alpha,\alpha/2}$ is the parabolic Hölder seminorm, and the last sum is omitted when $m = 0$. The following Sobolev embedding will be used below.

Lemma 4.2 (Parabolic Sobolev embedding; cf. [8]). *For any $m \in \mathbb{N}$, $n + 2 < p < \infty$ and $\alpha = 1 - (n+2)/p$, we have*

$$W_p^{2m,m}(\Omega) \hookrightarrow C^{2m-1+\alpha,\frac{2m-1+\alpha}{2}}(\Omega). \tag{4.10}$$

In particular, there is a constant $C = C(n, m, p) > 0$ such that

$$\|u\|_{C^{2m-1+\alpha,\frac{2m-1+\alpha}{2}}(\Omega)} \le C\|u\|_{W_p^{2m,m}(\Omega)}. \tag{4.11}$$

Lemma 4.3 (Interior parabolic estimates; cf. [8]). *Let u be a vector-valued classical solution of*

$$\partial_t u - a^{ij} D_{ij} u - b^i D_i u - cu = f \qquad \text{in } \Omega.$$

Assume that (a^{ij}) is uniformly elliptic, acts componentwise, and has a fixed modulus of continuity, and that b^i and c are bounded matrix-valued coefficients. If $Q' \Subset Q \subset \Omega$ are nested parabolic cylinders, away from the initial time and lateral boundary, then, for $1 < p < \infty$,

$$\|u\|_{W_p^{2,1}(Q')} \le C\big(\|u\|_{L^p(Q)} + \|f\|_{L^p(Q)}\big).$$

The constant depends on the dimensions, p, the ellipticity and coefficient bounds, the modulus of continuity of a, and the separation of the cylinders.

Proposition 4.4. *Fix $0 < \alpha < 1$ and choose $p > n+2$ so that $1-(n+2)/p > \alpha$. Suppose that harmonic coordinates and a Coulomb frame on $B = B_2(0)$ satisfy the uniform $W^{2,p}$ bounds established above. If U solves $(\partial_t - \Delta^E)U = 0$ on Ω, then*

$$\|U\|_{C^{2+\alpha,(2+\alpha)/2}(\Omega')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.12}$$

The norm on the left is the coordinate norm of the components in this frame. The constant depends on n, m, p and the stated coefficient bounds.

Proof. Write $U = e_\alpha u^\alpha$. We prove the stronger interior estimate

$$\|u\|_{W^{4,2}_p(\Omega')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.13}$$

The parabolic Sobolev embedding then gives the conclusion. We use a finite sequence of nested cylinders between Ω' and Ω; all norms below are taken on successively smaller cylinders. The notation $Q' \Subset Q$ records this interior restriction.

By the harmonic-coordinate and Coulomb conditions, equation (3.18) becomes

$$\partial_t u = \mathscr{L}u := a^{ij}D_{ij}u + b^i D_i u + cu, \qquad a^{ij} = g_0^{ij}, \quad b^j = 2g_0^{ij}\omega_i, \quad c = g_0^{ij}\omega_i\omega_j. \tag{4.14}$$

These coefficients are independent of time. Their $W^{2,p}$ norms are uniformly bounded, since $W^{2,p}$ is an algebra for $p > n$. In particular, a, b, c and their first derivatives are bounded. Their second derivatives need only belong to L^p. All constants in the following argument depend on these $W^{2,p}$ bounds, and not on any higher coordinate derivatives of the coefficients.

Step 1. First spatial derivatives. Lemma 4.3 gives

$$\|u\|_{W^{2,1}_p(Q')} \le C\|u\|_{L^p(Q)} \le C\|U\|_{L^\infty(\Omega)}, \tag{4.15}$$

$$\|u_t\|_{L^p(Q')} + \|Du\|_{L^p(Q')} + \|D^2u\|_{L^p(Q')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.16}$$

Since $p > n+2$, the parabolic Sobolev embedding applied to u already gives

$$\|u\|_{L^\infty(Q')} + \|Du\|_{L^\infty(Q')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.17}$$

This is the first-derivative estimate used for K_E. For $v_k = D_k u$, differentiating the full equation, including its lower-order terms, yields

$$(\partial_t - \mathscr{L})v_k = (D_k a^{ij})D_{ij}u + (D_k b^i)D_i u + (D_k c)u. \tag{4.18}$$

All terms on the right belong to L^p with a uniform bound. Consequently,

$$\|v_k\|_{W^{2,1}_p(Q')} \le C\big(\|v_k\|_{L^p(Q)} + \|D^2u\|_{L^p(Q)} + \|Du\|_{L^p(Q)} + \|u\|_{L^p(Q)}\big), \tag{4.19}$$

$$\|D^3u\|_{L^p(Q')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.20}$$

Applying the parabolic Sobolev embedding to v_k also gives

$$\|D^2u\|_{L^\infty(Q')} + \|D^2u\|_{C^{\alpha,\alpha/2}(Q')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.21}$$

Step 2. Time derivatives. Since all coefficients are independent of time, $w = u_t$ satisfies $(\partial_t - \mathscr{L})w = 0$. Its L^p norm is bounded by (4.16). Applying the same interior estimate to w gives

$$\|u_{tt}\|_{L^p(Q')} + \|Du_t\|_{L^p(Q')} + \|D^2u_t\|_{L^p(Q')} \le C\|U\|_{L^\infty(\Omega)}, \tag{4.22}$$

$$\|\partial_t D^2 u\|_{L^p(Q')} \le C\|U\|_{L^\infty(\Omega)}. \tag{4.23}$$

This argument uses the parabolic estimate for the full system; no scalar De Giorgi–Nash–Moser assertion for arbitrary systems is needed.

Step 3. Second spatial derivatives. Set $z_{kl} = D_k D_l u$. A second spatial differentiation gives

$$\begin{aligned}(\partial_t - \mathscr{L}) z_{kl} = {} & (D_k a^{ij}) D_{ij} D_l u + (D_l a^{ij}) D_{ij} D_k u + (D_k D_l a^{ij}) D_{ij} u \\ & + (D_k b^i) D_i D_l u + (D_l b^i) D_i D_k u + (D_k D_l b^i) D_i u \\ & + (D_k c) D_l u + (D_l c) D_k u + (D_k D_l c) u.\end{aligned}$$

The first line is controlled in L^p by (4.20), (4.21), and the $W^{2,p}$ bound for a. The remaining lines are controlled by the $W^{2,p}$ bounds for b, c and the bounds for u, Du, D^2u obtained in Step 1. Applying Lemma 4.3 once more yields

$$\|D^4 u\|_{L^p(\Omega')} \le C \|U\|_{L^\infty(\Omega)}, \tag{4.24}$$

$$\|u_{tt}\|_{L^p(\Omega')} \le C \|U\|_{L^\infty(\Omega)}. \tag{4.25}$$

Together with (4.16)–(4.23), these estimates prove (4.13).

Step 4. The time derivative of the spatial Hessian. The $W^{2,1}_p$ estimate for $w = u_t$ in Step 2 and Sobolev embedding give $\|w\|_\infty + \|Dw\|_\infty \le C\|U\|_\infty$ on a smaller cylinder. Differentiate its equation once in space:

$$(\partial_t - \mathscr{L}) D_k w = (D_k a^{ij}) D_{ij} w + (D_k b^i) D_i w + (D_k c) w.$$

The right-hand side is bounded in L^p by Step 2 and the bounded first derivatives of a, b, c. The interior estimate for $D_k w$, followed by Sobolev embedding, therefore yields

$$\|u_t\|_{L^\infty(Q')} + \|D^2 u_t\|_{L^\infty(Q')} \le C \|U\|_{L^\infty(\Omega)}. \tag{4.26}$$

One may choose the finitely many intermediate cylinders in advance so that the final cylinder contains Ω'. This argument differentiates only the fixed coefficients of Δ^E; it does not use coefficients of $L_0 - \Delta^E$. □

Lemma 4.5. *Suppose U is a smooth section on B, expressed in the harmonic coordinates and Coulomb frame above. Then*

$$\sup_{x \in B'} \left\{ |\nabla U|_g(x) + |\nabla\nabla U|_g(x) + [\nabla\nabla U]^{(\frac{1}{2})}_{g,\alpha}(x) \right\} \le C \|U\|_{C^{2+\alpha}(B)}. \tag{4.27}$$

Notice that (cf. Definition 4.1)

$$[\nabla\nabla U]^{(\frac{1}{2})}_{g,\alpha}(x) := \sup_{y \in B_{\frac{1}{2}}(x)} \frac{|U_{ij}(x) - \tau_\gamma(x,y) U_{ij}(y)|}{|\gamma|^\alpha}, \tag{4.28}$$

where γ is any shortest geodesic connecting x and y under the metric g, $\tau_\gamma(x,y)$ is the parallel transport from y to x along the geodesic γ.

Proof. In the chosen frame, the first two covariant derivatives are linear combinations of u, Du, D^2u, with coefficients formed from $\omega, D\omega$ and the Christoffel symbols. Their required Hölder norms are bounded by the $W^{2,p}$ bounds for g_0 and ω, with p chosen sufficiently large. Explicitly,

$$\begin{aligned}(\nabla^{E,2} U)_{ij} = e\big(& D_{ij} u + \omega_i D_j u + \omega_j D_i u - \Gamma^k_{ij} D_k u \\ & + (D_i \omega_j + \omega_i \omega_j - \Gamma^k_{ij} \omega_k) u \big).\end{aligned}$$

Here $\Gamma, \omega, D\omega$ have uniform C^α bounds because $1 - n/p > \alpha$. No derivative of Γ is needed in this conversion. Parallel transport along a short minimizing geodesic solves a linear ordinary differential equation with uniformly bounded coefficients; its difference from the coordinate identification is bounded by a constant times the length of that geodesic. Combining this observation with the coordinate Hölder estimate proves (4.27). The metric comparison ensures that the short geodesics used here stay inside B. □

Combining Proposition 4.4 and Lemma 4.5, we obtain the following corollary.

Corollary 4.6. *Suppose U satisfies the heat equation. Then we have*

$$\sup_{\Omega'}\big(|\nabla U|_g + |\nabla\nabla U|_g + [\nabla\nabla U]^{(\frac{1}{2})}_{g,\alpha} + |\partial_t U| + |\partial_t \nabla^{E,2} U|\big) \le C\|U\|_{L^\infty(\Omega)}. \tag{4.29}$$

For the last term, apply the preceding coordinate-to-covariant formula to u_t and use (4.26). The connection and metric are fixed in time, so no coefficient is differentiated in t.

4.1. **Heat kernel estimates.** We now apply the local estimates to the rough-Laplacian heat kernel on $E = F_0^* TN$. All distances, volumes, and covariant derivatives in this subsection refer to the fixed induced metric g_0. We write $V_x(r) = \mathrm{Vol}_{g_0}(B_r(x))$. The pullback to a tangent ball is used only for local estimates of solutions; it need not be an injective map onto its image in M.

Lemma 4.7. *Let (M, g_0) be complete with $|\operatorname{Rm}_{g_0}| \le \Lambda_0$, and let E be a Euclidean vector bundle with a metric connection. The heat kernel K_E of Δ^E satisfies*

$$|K_E(x, y, t)| \le C V_x(\sqrt{t})^{-1} \exp\left(-\frac{d(x, y)^2}{Ct}\right), \qquad 0 < t \le 1, \tag{4.30}$$

where C depends on n, Λ_0 and the bundle rank.

Proof. The Kato inequality gives semigroup domination $|e^{t\Delta^E} s| \le e^{t\Delta}|s|$. Thus the operator norm of the bundle kernel is bounded by the scalar heat kernel: test the domination inequality against approximations of a delta section at y, with an arbitrary unit vector in E_y. If a Hilbert–Schmidt norm is used instead, the bundle rank changes only the constant. The scalar Gaussian estimate under the lower Ricci bound gives

$$H(x, y, t) \le \frac{Ce^{Ct}}{\sqrt{V_x(\sqrt{t})V_y(\sqrt{t})}} \exp\left(-\frac{d(x, y)^2}{Ct}\right).$$

To pass to the one-sided volume factor, put $R = \sqrt{t}$. The inclusion $B_R(x) \subset B_{R+d(x,y)}(y)$ and Bishop–Gromov comparison imply

$$\frac{V_x(R)}{V_y(R)} \le C\left(1 + \frac{d(x, y)}{R}\right)^n e^{C(R+d(x,y))}.$$

Since $R \le 1$, both this polynomial factor and the extra exponential are absorbed by increasing the constant in the Gaussian denominator. This proves (4.30). No positive lower bound for $V_x(R)$ is used; the actual volume remains in the estimate. □

Fix $x, y \in M$ and $t \in (0, 1]$. Choose a positive constant ϑ, depending only on the geometric bounds and the exponent used in the local estimates, and put $r = \vartheta\sqrt{t}$. We choose ϑ sufficiently small that $4r^2 < t/2$, the rescaled base curvature is below the threshold in Lemma 3.4, and the pullback connection satisfies the small-curvature hypothesis in Lemma 3.5. The constants below are independent of x, y, t. Here the required bounds before rescaling are precisely

$$|\operatorname{Rm}_{g_0}| \le C(n)(K_0 + \Lambda^2), \qquad |R^E| \le CK_0, \qquad |\nabla R^E| \le C(K_1 + \Lambda K_0).$$

With base metric $r^{-2}g_0$, these three bounds acquire the factors r^2, r^2, r^3, respectively. We can therefore choose $\vartheta > 0$ depending only on $n, m, K_0, K_1, \Lambda, p$, so that all the rescaled metric and connection estimates used below have fixed constants.

For $w \in B_{100r}(x)$ and $t - 4r^2 \le s \le t$, Gaussian decay and volume comparison imply

$$\sup_{\substack{w\in B_{100r}(x)\\ t-4r^2\le s\le t}} |K_E(w, y, s)| \le CV_x(\sqrt{t})^{-1} \exp\left(-\frac{d(x, y)^2}{Ct}\right) =: \mathcal{K}_{x,y,t}. \tag{4.31}$$

Indeed, $s \in [t/2, t]$ and

$$d(w, y)^2 \ge \tfrac{1}{2}d(x, y)^2 - d(w, x)^2.$$

The resulting factor $\exp(Cr^2/t)$ is uniformly bounded, and the volumes centered at w and x are comparable at radius $\sqrt{t}$.

Equip M with the rescaled metric $\widetilde{g} = r^{-2}g_0$ and consider

$$\varphi(v) = \operatorname{Exp}_x^{\widetilde{g}}(v).$$

On a sufficiently large fixed tangent ball, $\widehat{g} = \varphi^*\widetilde{g}$ is smooth and has a uniformly positive harmonic radius at 0. More precisely, use the tangent ball of radius $100n\pi$ in Lemma 3.4. The curvature bound excludes conjugate points there. The Gauss lemma identifies the radial distance from 0 with the norm of the tangent vector, which gives the injectivity-radius bound needed for harmonic coordinates on the pulled-back ball. This assertion concerns the lifted metric, regardless of how many points the map φ identifies in M. Choose harmonic coordinates ψ, centred at 0, and restrict them to the radius-10 metric ball. The larger harmonic chart supplied by the lemma leaves room for all subsequent interior restrictions. The metric comparison gives

$$B_{5\sqrt{2}}(0) \subset \psi(B_{10}^{\widehat{g}}(0)) \subset B_{10\sqrt{2}}(0). \tag{4.32}$$

Apply Lemma 3.5 on a slightly larger coordinate ball and restrict to $B_2(0)$ to obtain the required Coulomb frame. Equivalently, its proof applies to the lifted connection with the rescaled curvature bounds just displayed. In these coordinates the metric coefficients and the connection form have uniformly bounded $W^{2,p}$ norms. These are the only coefficient bounds entering Proposition 4.4.

For fixed y, view the heat kernel as a section of $E \otimes E_y^*$ in the first variable, and pull it back to this frame:

$$\overline{K}(z, \sigma) = K_E\big(\varphi(\psi^{-1}(z)), y, t + r^2\sigma\big), \qquad -4 \le \sigma \le 0.$$

The displayed equality is understood after the indicated bundle identification. Since φ is a local isometry for the pullback metric, $\overline{K}$ solves the corresponding bundle heat equation. No integration over the fibers of φ is used. In particular,

$$\|\overline{K}\|_{L^\infty(\Omega)} \le \mathcal{K}_{x,y,t}. \tag{4.33}$$

The image of the coordinate cylinder lies in $B_{100r}(x) \times [t-4r^2, t]$, where (4.31) applies. The local L^p norms in the parabolic estimates are bounded by this L^∞ norm times the fixed Euclidean cylinder volume. We never compare such an integral with the integral of the kernel over its possibly multiply covered image in M. Proposition 4.4 gives

$$\|\overline{K}\|_{C^{2+\alpha,(2+\alpha)/2}(\Omega')} \le C\mathcal{K}_{x,y,t}. \tag{4.34}$$

Applying the same interior argument to $\partial_\sigma \overline{K}$, as detailed in (4.26), also controls its spatial Hessian. In order, the $W^{2,1}_p$ estimate for $\overline{K}$ controls its first spatial derivative; the differentiated equation for $D\overline{K}$ controls $D^2\overline{K}$ and its Hölder seminorm; and the equations for $\partial_\sigma \overline{K}$ and $D\partial_\sigma \overline{K}$ control the time derivative of the Hessian. Passing from coordinate derivatives to covariant derivatives therefore yields

$$\begin{aligned}\sup_{\Omega''}\big(&|\overline{K}| + |\widehat{\nabla}\overline{K}| + |\widehat{\nabla}^2\overline{K}| + [\widehat{\nabla}^2\overline{K}]^{(c)}_\alpha \\ &+ |\partial_\sigma \overline{K}| + |\partial_\sigma \widehat{\nabla}^2\overline{K}|\big) \le C\mathcal{K}_{x,y,t},\end{aligned} \tag{4.35}$$

where Ω'' is a fixed cylinder containing $(0,0)$ and $c>0$ is fixed. Here the coefficients are time independent, so covariant differentiation commutes with ∂_σ.

Every minimizing geodesic from x of sufficiently small length lifts to a radial geodesic in the tangent ball: a geodesic with initial velocity v lifts as $s \mapsto sv$. Parallel transport of the tensor kernel commutes with this local isometry and with the lifted bundle connection. The local estimate therefore applies to every such geodesic separately, with the same constant. Taking their supremum gives exactly the seminorm in Definition 4.1, even at points joined to x by several minimizing geodesics. This is the step that descends the Hölder estimate to the possibly collapsed base manifold.

Under $\widetilde{g} = r^{-2}g_0$, the norms of the first and second covariant derivatives acquire the factors r and r^2. The spatial Hölder quotient for the Hessian acquires $r^{2+\alpha}$, while $\partial_\sigma = r^2\partial_t$. Consequently, scaling back gives at (x,y,t)

$$\begin{aligned}&|K_E| + r|\nabla_x K_E| + r^2|\nabla_x^2 K_E| + r^2|\partial_t K_E| \\ &\quad + r^{2+\alpha}[\nabla_x^2 K_E]^{(cr)}_\alpha + r^4|\partial_t \nabla_x^2 K_E| \le C\mathcal{K}_{x,y,t}.\end{aligned} \tag{4.36}$$

This proves the following theorem.

Theorem 4.8. *Let $F_0 : M^n \to (N^{n+m}, h)$ be a complete isometric immersion satisfying*

$$\sup_N |\operatorname{Rm}_N| \le K_0, \qquad \sup_N |\nabla \operatorname{Rm}_N| \le K_1, \qquad \sup_M |A_0| \le \Lambda.$$

Let $K_E(x,y,t) : E_y \to E_x$ be the heat kernel of Δ^E on $E = F_0^*TN$. For every $0<\alpha<1$, there are constants $c_*, C > 0$, depending only on $n, m, K_0, K_1, \Lambda, \alpha$, such that, with

$$\rho_t = c_*\sqrt{t}, \tag{4.37}$$

we have

$$\begin{aligned}&|K_E| + t^{1/2}|\nabla_x K_E| + t|\nabla_x^2 K_E| + t|\partial_t K_E| \\ &\quad + t^{1+\alpha/2}[\nabla_x^2 K_E]^{(\rho_t)}_\alpha + t^2|\partial_t \nabla_x^2 K_E| \\ &\quad \le CV_x(\sqrt{t})^{-1}\exp\left(-\frac{d(x,y)^2}{Ct}\right)\end{aligned} \tag{4.38}$$

for $x, y \in M$ and $0 < t \leq 1$. The Hölder seminorm is taken in the first variable, using the induced connection on $T^{*\otimes 2}M \otimes E$ and keeping the source fiber E_y fixed. The constants are independent of the injectivity radii and of any positive lower bound for the volumes of metric balls. The only volume factor is the actual $V_x(\sqrt{t})$ appearing in the estimate.

In particular, all second-order kernel bounds used in the next subsection concern Δ^E. The bounded coefficients of $L_0 - \Delta^E$ will be estimated as part of the forcing term in the weighted spaces, without differentiating them.

4.2. **The Calderón–Zygmund inequality.** To apply the abstract Calderón–Zygmund theorem of [7], one needs both the kernel estimates and an L^2 operator estimate. The latter does not follow from a pointwise bound for the kernel alone. For compactly supported smooth f, set

$$u(x,t) = \int_0^t \int_M K_E(x, y, t-s) f(y,s)\, d\mu(y)\, ds.$$

Then $(\partial_t - \Delta^E)u = f$ and $u(\cdot, 0) = 0$. Let $\mathsf{A} = -\Delta^E$ denote its nonnegative self-adjoint realization on $L^2(E)$. Expanding $f = u_t + \mathsf{A}u$ and integrating in time gives

$$\int_0^1 \left(\|u_t\|_2^2 + \|\mathsf{A}u\|_2^2 \right) dt + \|\nabla^E u(1)\|_2^2 = \int_0^1 \|f\|_2^2\, dt. \tag{4.39}$$

Indeed, the cross term is $2\langle u_t, \mathsf{A}u \rangle = \partial_t \|\nabla^E u\|_2^2$, and the initial value is zero. Spectral regularisation justifies the identity before any second-spatial-derivative estimate is known. The L^2 contraction of $e^{t\Delta^E}$ and the integral formula for u also give $\|u\|_{L^2(M\times(0,1))} \leq \|f\|_{L^2(M\times(0,1))}$. Finally, $\|\nabla^E u\|_{L^2}^2 \leq \|\mathsf{A}u\|_{L^2} \|u\|_{L^2}$. It follows that

$$\int_0^1 \int_M \left(|u|^2 + |\nabla^E u|^2 + |\Delta^E u|^2 \right) \leq C \int_0^1 \int_M |f|^2. \tag{4.40}$$

To pass from the Laplacian to the full Hessian, use the commutator

$$\Delta^{T^*M\otimes E}(\nabla^E u) - \nabla^E \Delta^E u = \mathrm{Rm}_M * \nabla^E u + R^E * \nabla^E u + (\nabla R^E) * u. \tag{4.41}$$

Thus no derivative of Rm_M, and hence no derivative of A_0 through the Gauss equation, occurs. Integrating the equation against $\nabla^E u$ gives, first for compactly supported smooth sections,

$$\begin{aligned} \int_M |\nabla^{E,2} u|^2 \leq & \int_M |\Delta^E u|^2 + C \int_M (|\,\mathrm{Rm}_M\,| + |R^E|) |\nabla^E u|^2 \\ & + C \int_M |\nabla R^E|\, |u|\, |\nabla^E u|. \end{aligned}$$

On the complete manifold, compactly supported smooth sections form a graph-norm core for the rough Laplacian. Applying this inequality to such approximations extends it to $u(t)$; the inequality itself shows that their Hessians converge in L^2. This also justifies the spatial integrations by parts without assuming a lower injectivity-radius bound. The bounds (2.3) and (2.6), together with (4.40), imply

$$\int_0^1 \int_M |\nabla^{E,2} u|^2 \leq C \int_0^1 \int_M |f|^2. \tag{4.42}$$

Thus $\mathcal{T}f = \nabla^{E,2}u$ is an L^2-bounded singular integral operator. Its kernel away from the parabolic diagonal is

$$\mathcal{K}(x,t;y,s) = \nabla_x^{E,2} K_E(x,y,t-s), \qquad s<t.$$

Write $\theta = t-s$. The three kernel hypotheses in [7] are precisely supplied by (4.38):

$$\begin{aligned} &\theta|\mathcal{K}| + \theta^{1+\alpha/2}[\mathcal{K}]_\alpha^{(c_*\sqrt{\theta})} + \theta^2|\partial_t\mathcal{K}| \\ &\qquad \le CV_x(\sqrt{\theta})^{-1}\exp\left(-\frac{d(x,y)^2}{C\theta}\right), \qquad 0<\theta\le 1. \end{aligned} \tag{4.43}$$

The time derivative here is the output-time derivative with y, s fixed. If a larger fixed comparison radius $c_0\sqrt{\theta}$ is used in the abstract theorem, the same Hölder estimate holds after enlarging C. For distances at most $c_*\sqrt{\theta}$ this is already (4.43). For distances between $c_*\sqrt{\theta}$ and $c_0\sqrt{\theta}$, use the size bound at both endpoints, volume comparison, and the isometry of parallel transport; division by the distance to the power α contributes at most $(c_*\sqrt{\theta})^{-\alpha}$. The singular integral is understood as the distributional Hessian of the heat potential, or equivalently by smooth approximation in L^2; absolute integration of its size bound at $s=t$ is not asserted. The abstract theorem is formulated for kernels between smooth metric vector bundles. Here the input bundle is E and the output bundle is $T^*M\otimes T^*M\otimes E$, with its induced connection. The Hölder seminorm is defined by parallel transport, as in (4.38), so the theorem applies without choosing a global trivialization. The abstract Calderón–Zygmund theorem of [7, Theorem 6.4] therefore yields the corresponding global L^p estimate on an interior time interval, for $2\le p<\infty$.

Proposition 4.9 (Zero-initial-value estimate)**.** *Under the hypotheses of Theorem 4.8, let $2\le p<\infty$ and $0<T\le 1/4$. For $f\in L^p(M\times(0,T);E)$, its heat potential u satisfies*

$$\|\nabla^{E,2}u\|_{L^p(M\times(0,T))} + \|\partial_t u\|_{L^p(M\times(0,T))} \le C\|f\|_{L^p(M\times(0,T))}, \tag{4.44}$$

where $C = C(n,m,p,K_0,K_1,\Lambda)$ is independent of T.

Proof. It suffices first to take f smooth and compactly supported in $M\times(0,T)$. Extend f and its zero-initial-value heat potential by zero to negative times. The same causal operator can then be considered on the interval $(-3T,T)$, whose last quarter is $(0,T)$. Apply the interior-time global Calderón–Zygmund theorem just established on this larger interval. To check that its constant is independent of T, rescale space and time by

$$g' = (4T)^{-1}g_0, \qquad t' = \frac{t-T}{4T}.$$

The interval becomes $(-1,0)$, and the output interval becomes $(-1/4,0)$. The bounds for $\mathrm{Rm}_M, R^E, \nabla R^E$ acquire the factors $4T, 4T, (4T)^{3/2}$, respectively, so they remain uniformly bounded. The kernel bounds (4.43) and the L^2 operator bound therefore have uniform constants after scaling. Both the Hessian and the forcing term acquire the same factor $4T$, and the common space-time volume factor cancels in their L^p norms. This proves the Hessian estimate on the full interval $(0,T)$.

The equation gives $|u_t| \le \sqrt{n}\,|\nabla^{E,2}u| + |f|$, proving the other estimate. Finally, the heat semigroup is an L^p contraction by domination, so $\|u\|_{L^p(M\times(0,T))} \le T\|f\|_{L^p(M\times(0,T))}$. Density and (4.44) extend the conclusion to every $f\in L^p$, with the Hessian and time derivative understood weakly. □

Corollary 4.10. *Let $F_0 : M^n \to (N^{n+m}, h)$ be a complete isometric immersion with*

$$\sup_N |\operatorname{Rm}_N| \le K_0, \qquad \sup_N |\nabla \operatorname{Rm}_N| \le K_1, \qquad \sup_M |A_0| \le \Lambda.$$

Let $2 \le p < \infty$, and suppose that smooth sections u, f of $E = F_0^* TN$ satisfy

$$(\partial_t - \Delta^E)u = f \quad \text{on } M \times [0,1], \qquad u, f \in L^p(M \times [0,1]).$$

Then

$$\int_{3/4}^1 \int_M \left(|\nabla^{E,2} u|^p + |\partial_t u|^p\right) d\mu\, dt \le C \int_0^1 \int_M \left(|u|^p + |f|^p\right) d\mu\, dt, \tag{4.45}$$

where $C = C(n, m, p, K_0, K_1, \Lambda)$.

Proof. Choose $\eta \in C^\infty([0,1])$ with $\eta = 0$ on $[0, 1/4]$, $\eta = 1$ on $[1/2, 1]$, and $|\eta'| \le C$. Set $v = \eta u$. Then

$$(\partial_t - \Delta^E)v = \eta f + \eta' u =: F, \qquad v(\cdot, 0) = 0.$$

The L^p uniqueness and Duhamel formula for the connection heat semigroup identify v with the heat potential of F. Applying the global operator estimate established above gives

$$\|\nabla^{E,2} u\|_{L^p(M\times[3/4,1])} \le C\|F\|_{L^p(M\times[0,1])} \le C\big(\|u\|_{L^p(M\times[0,1])} + \|f\|_{L^p(M\times[0,1])}\big).$$

Finally, $|\partial_t u| \le \sqrt{n}\,|\nabla^{E,2} u| + |f|$ proves the time-derivative estimate. No spatial localization is used. □

5. A priori estimates in weighted spaces

Fix $p = n + 4 > n + 2$. For $0 < t \le T \le 1/4$ and $0 < r \le \min\{1, \sqrt{t}\}$, define the backward cylinder

$$Q_r(x, t) := B_{g_0}(x, r) \times (t - r^2/2, t). \tag{5.1}$$

For an E-valued tensor f, put

$$\|f\|_{\mathcal{Y}_T} := \sup_{x,t,r} r \left(\frac{1}{r^2 V_x(r)} \int_{Q_r(x,t)} |f|^p \, \mathrm{d}\mu_{g_0}\, \mathrm{d}s\right)^{1/p}. \tag{5.2}$$

For a section u with $u(\cdot, 0) = 0$, put

$$\begin{aligned}\|u\|_{\mathcal{X}_T} := {} & \|u\|_{L^\infty(M\times(0,T))} + \|\nabla^E u\|_{L^\infty(M\times(0,T))} \\ & + \sup_{x,t,r} r \left(\frac{1}{r^2 V_x(r)} \int_{Q_r(x,t)} |\nabla^{E,2} u|^p \, \mathrm{d}\mu_{g_0}\, \mathrm{d}s\right)^{1/p}.\end{aligned} \tag{5.3}$$

The supremum in both definitions is over the triples allowed in (5.1). We take $\mathcal{Y}_T$ to be the space of measurable tensors with finite (5.2), modulo equality almost everywhere. The space $\mathcal{X}_T$ consists of measurable sections, modulo equality almost everywhere, with weak first and second spatial covariant derivatives, finite (5.3), and zero uniform initial trace, by which we mean

$$\lim_{\tau \downarrow 0} \|u\|_{L^\infty(M\times(0,\tau))} = 0.$$

Both spaces are Banach spaces. Indeed, a Cauchy sequence in $\mathcal{Y}_T$ converges in L^p on each admissible cylinder, and the resulting local limits agree and converge in the defining norm. For $\mathcal{X}_T$, the L^∞ limits of the sections and their first derivatives and the $\mathcal{Y}_T$ limit of their

Hessians agree as distributional derivatives. The zero uniform initial trace is preserved by L^∞ convergence. In particular, if f is bounded, then

$$\|f\|_{\mathcal{Y}_T} \le \sqrt{T}\,\|f\|_{L^\infty}. \tag{5.4}$$

Proposition 5.1. *Assume the normalised geometric bounds in Section 2 and the heat-kernel estimates of Theorem 4.8, and let $0 < T \le 1/4$. Let K_E denote the heat kernel of $\partial_t - \Delta^E$, and define*

$$(\mathcal{H}f)(x,t) := \int_0^t \int_M K_E(x,t;y,s) f(y,s)\ \mathrm{d}\mu_{g_0}(y)\,\mathrm{d}s. \tag{5.5}$$

Then

$$\|\mathcal{H}f\|_{\mathcal{X}_T} \le C(n,m)\|f\|_{\mathcal{Y}_T}. \tag{5.6}$$

Proof. We divide the proof into four steps. Throughout the proof, $Y = \|f\|_{\mathcal{Y}_T}$. We first consider smooth sources supported away from $t = 0$; the estimates below are independent of this restriction.

Step 1. The contribution of the terminal cylinder. Let $p' = p/(p-1)$. By (4.38), volume comparison, and a decomposition into spatial annuli, we have

$$\int_M \exp\left(-\frac{p' d(x,y)^2}{C\theta}\right) \mathrm{d}\mu(y) \le C V_x(\sqrt{\theta}), \qquad 0 < \theta \le 1. \tag{5.7}$$

Indeed, the annulus of radii $2^k\sqrt{\theta}$ and $2^{k+1}\sqrt{\theta}$ contributes at most $C2^{n(k+1)}\exp(C2^k\sqrt{\theta} - c4^k)V_x(\sqrt{\theta})$. The exponential volume-growth factor is needed when the outer radius exceeds one, and the resulting series converges uniformly for $\theta \le 1$.

Fix (x,t) and put $R = \sqrt{t}$. For $j = 0, 1$, we obtain

$$\begin{aligned}\int_0^{R^2} \int_M |\nabla_x^j K_E(x,R^2;y,R^2-\theta)|^{p'}\ \mathrm{d}\mu(y)\,\mathrm{d}\theta \\ \le C\int_0^{R^2} \theta^{-jp'/2} V_x(\sqrt{\theta})^{1-p'}\,\mathrm{d}\theta.\end{aligned} \tag{5.8}$$

Volume comparison gives

$$V_x(\sqrt{\theta})^{1-p'} \le C V_x(R)^{1-p'}(R/\sqrt{\theta})^{n(p'-1)}.$$

Since $jp' + n(p'-1) < 2$ for $j = 0, 1$ and $p = n + 4$, it follows that

$$\left(\int_0^{R^2} \int_M |\nabla_x^j K_E|^{p'}\right)^{1/p'} \le C V_x(R)^{-1/p} R^{2/p' - j}. \tag{5.9}$$

The source norm, applied to the actual cylinder $Q_R(x,t) = B(x,R) \times (t/2, t)$, gives

$$\|f\|_{L^p(Q_R(x,t))} \le R^{-1}(R^2 V_x(R))^{1/p} Y. \tag{5.10}$$

Consequently, the contribution of this terminal cylinder satisfies

$$|\mathcal{H}(f\mathbf{1}_{Q_R})(x,t)| \le CRY, \qquad |\nabla\mathcal{H}(f\mathbf{1}_{Q_R})(x,t)| \le CY. \tag{5.11}$$

Notice that this estimate does not include the earlier interval $(0, t/2)$.

Step 2. The initial-time contribution and spatial tails. We first record an L^1 estimate which remains valid at the initial time. For $\sigma_\ell = 2^{-\ell}t$, cover $B(z,R)$ by balls of radius

$r_\ell = \sqrt{\sigma_\ell}$ with bounded overlap. Since $r_\ell \le R \le 1$, volume comparison bounds the sum of their volumes by $CV_z(R)$. Applying the $\mathcal{Y}_T$ bound on the cylinders ending at σ_ℓ yields

$$\int_{\sigma_\ell/2}^{\sigma_\ell} \int_{B(z,R)} |f| \le C\sqrt{\sigma_\ell}\, V_z(R) Y.$$

Summing this geometric series gives

$$\int_0^t \int_{B(z,R)} |f| \le CRV_z(R)Y, \qquad R = \sqrt{t}. \tag{5.12}$$

This argument uses only L^1 integrability at $t = 0$; the $\mathcal{Y}_T$ norm does not in general imply L^p integrability on a cylinder meeting that time.

On $B(x, R) \times (0, t/2)$ we have $t - s \ge t/2$, so the kernel estimates and (5.12) bound the contribution of this region by $CR^{1-j}Y$, for $j = 0, 1$. For the remaining spatial region, put

$$A_k = B(x, 2^{k+1}R) \setminus B(x, 2^k R), \qquad k \ge 0.$$

A covering of A_k by balls $B(z_a, R)$ of bounded overlap satisfies

$$\sum_a V_{z_a}(R) \le CV_x(2^{k+3}R). \tag{5.13}$$

For $0 < \theta = t - s \le R^2$, set $a = R^2/\theta \ge 1$. The Gaussian bound gives

$$|\nabla_x^j K_E(x, t; y, s)| \le \frac{CR^{-j}}{V_x(R)} a^{(n+j)/2} e^{-c4^k a} \le \frac{CR^{-j}}{V_x(R)} e^{-c'4^k}, \qquad y \in A_k. \tag{5.14}$$

Combining (5.12)–(5.14), we obtain

$$\begin{aligned} \int_0^t \int_{A_k} |\nabla_x^j K_E(x, t; y, s)|\, |f(y, s)|\, \mathrm{d}\mu(y)\, \mathrm{d}s \\ \le CR^{1-j} 2^{nk} \exp(C2^k R - c'4^k) Y. \end{aligned} \tag{5.15}$$

The sum over k is uniformly bounded. Together with Step 1 and the early-time estimate, this proves

$$\|\mathcal{H}f\|_{L^\infty} \le C\sqrt{T}\, Y, \qquad \|\nabla^E \mathcal{H}f\|_{L^\infty} \le CY. \tag{5.16}$$

This is the volume-normalised near/far decomposition used in [5, Proposition 4.2] and [1, Proposition 4.1], with a separate decomposition of the source near the initial time.

Step 3. The global Calderón–Zygmund estimate. The direct kernel calculation above does not apply to two spatial derivatives: the corresponding time integral is divergent. We instead use the zero-initial-value estimate of Proposition 4.9 for Δ^E. We recall its energy and initial-time arguments here to make the application of the Calderón–Zygmund theorem from [7] explicit. First, let $(\partial_t - \Delta^E)v = f$ with $v(\cdot, 0) = 0$. Multiplication by $\partial_t v$ and integration give

$$\int_0^T \int_M |\partial_t v|^2 + \int_M |\nabla^E v(\cdot, T)|^2 \le \int_0^T \int_M |f|^2. \tag{5.17}$$

Since $\Delta^E v = \partial_t v - f$, we have

$$\|\Delta^E v\|_{L^2(M\times(0,T))} \le 2\|f\|_{L^2(M\times(0,T))}. \tag{5.18}$$

The bundle Bochner identity, integration by parts, and Young's inequality imply

$$\|\nabla^{E,2} v\|_2^2 \le C\big(\|\Delta^E v\|_2^2 + \|\nabla^E v\|_2^2 + \|v\|_2^2\big). \tag{5.19}$$

The commutators involve $\mathrm{Rm}_M * \nabla v$, $R^E * \nabla v$, and $(\nabla R^E) * v$, whose coefficients are bounded by (2.3) and (2.6). The L^2 contraction of the connection heat semigroup and $T \le 1$ give $\|v\|_2 \le T\|f\|_2$; also, $\|\nabla^E v\|_2^2 \le \|v\|_2 \|\Delta^E v\|_2$. These inequalities prove the required L^2 operator bound.

The Gaussian size and off-diagonal Hölder bounds for $\nabla_x^{E,2} K_E$ are supplied by the preceding heat-kernel estimates. To obtain an estimate up to the initial time from the interior-time theorem, extend f and v by zero to negative times and apply that theorem on $M \times (-3T, T)$, after parabolic rescaling. The last quarter of this interval is $(0, T)$, and the rescaled geometric bounds remain controlled because $4T \le 1$. The resulting estimate is

$$\|\nabla^{E,2} v\|_{L^p(M\times(0,T))} \le C\big(\|f\|_{L^p(M\times(0,T))} + T^{-1}\|v\|_{L^p(M\times(0,T))}\big).$$

The L^p contraction of the connection heat semigroup and Young's inequality in time give $\|v\|_{L^p} \le T\|f\|_{L^p}$. The zero extension solves the equation distributionally across $t = 0$, since v has zero initial trace. Thus, for each fixed $p \ge 2$,

$$\|\partial_t v\|_{L^p} + \|\nabla^{E,2} v\|_{L^p} \le C(n, m, p)\|f\|_{L^p}. \tag{5.20}$$

Here and below all space-time norms in this formula are over $M \times (0, T)$.

Step 4. Localisation in the weighted norm. Fix $Q_r(x, t)$. Set

$$P_R = B(x, 2R) \times (\max\{0, t - 3R^2/4\}, t), \qquad f_{\mathrm{near}} = f\mathbf{1}_{P_r}, \qquad f_{\mathrm{far}} = f - f_{\mathrm{near}}.$$

Since $t \ge r^2$, the time interval of P_r starts at or after $r^2/4$. It can therefore be covered by a fixed number of admissible cylinders of radius comparable to r. A spatial covering and volume comparison give

$$\|f_{\mathrm{near}}\|_{L^p} \le Cr^{-1}(r^2 V_x(r))^{1/p} Y. \tag{5.21}$$

Applying (5.20) and restricting to $Q_r(x, t)$, we obtain

$$r\left(\fint_{Q_r(x,t)} (|\partial_t \mathcal{H} f_{\mathrm{near}}|^p + |\nabla^{E,2}\mathcal{H} f_{\mathrm{near}}|^p)\right)^{1/p} \le CY. \tag{5.22}$$

For the far contribution, put $R_k = 2^k r$ and $D_k = P_{R_k} \setminus P_{R_{k-1}}$, for $k \ge 1$. We first consider the indices up to the first k_* for which $R_{k_*} \ge \sqrt{t}$; in particular, $R_{k_*} < 2\sqrt{t}$ unless $r = \sqrt{t}$. The same time-layer argument as in Step 2 gives

$$\int_{D_k} |f| \le CR_k V_x(CR_k) Y. \tag{5.23}$$

For completeness, if the time interval stays a distance comparable to R_k^2 from zero, a finite covering by cylinders of radius comparable to R_k proves this estimate. If it reaches zero, divide it into the layers $(\sigma/2, \sigma)$ and use spatial balls of radius $\sqrt{\sigma}$; the resulting sum of $\sqrt{\sigma}$ is at most CR_k. All radii used here are at most a fixed dimensional multiple of one.

For $(z, \tau) \in Q_r(x, t)$ and $(y, s) \in D_k$ with $s < \tau$, either $d(z, y) \ge cR_k$ or $\tau - s \ge cR_k^2$. The second-derivative kernel bound consequently gives

$$|\nabla_z^{E,2} K_E(z, \tau; y, s)| \le \frac{C}{R_k^2 V_x(CR_k)}. \tag{5.24}$$

Thus each D_k contributes at most $CR_k^{-1}Y$, and

$$\sum_{k=1}^{k_*} CR_k^{-1}Y \le Cr^{-1}Y.$$

The complement of $P_{R_{k_*}}$ is handled by a further spatial annular decomposition at radius $\sqrt{t}$, rather than by a single annulus of finite volume. Sources in its spatially bounded part lie at times at most $t/4$; the rest are spatially separated from $Q_r(x,t)$. Using (5.12) and the argument of (5.15), now with $j = 2$, bounds this remaining contribution by $Ct^{-1/2}Y \le Cr^{-1}Y$. When $r = \sqrt{t}$, the same argument applies directly to the complement of P_r. We have proved

$$|\nabla^{E,2}\mathcal{H}f_{\mathrm{far}}| \le Cr^{-1}Y \qquad \text{on } Q_r(x,t). \tag{5.25}$$

Since f_{far} vanishes on the target cylinder, $\partial_t\mathcal{H}f_{\mathrm{far}} = \Delta^E\mathcal{H}f_{\mathrm{far}}$ there. Combining (5.22) and (5.25), and taking the supremum over the admissible cylinders, proves the weighted Hessian bound. Together with (5.16), this proves (5.6).

For a general measurable source in $\mathcal{Y}_T$, use spatial and temporal truncations. The preceding L^1 estimates and the Gaussian tails define the heat potential, and local weak compactness in $W_p^{2,1}$ away from $t = 0$ gives the same estimates by lower semicontinuity. Moreover, (5.16) on $(0,s)$ implies $\|\mathcal{H}f(\cdot,s)\|_\infty \le C\sqrt{s}\,Y$, so the initial trace is zero. No convergence of the truncations in the full $\mathcal{X}_T$ norm is asserted. □

6. Short-time existence of mean curvature flow

6.1. **The mean–DeTurck flow.** We work in the normalised scale, including the bound $|\nabla^{N,2}\operatorname{Rm}_h| \le 1$. The exact equation in Lemma 2.3 is

$$\partial_t U = \mathbf{H}_0 + L_0 U + \mathcal{Q}(U).$$

Here $\mathcal{Q}$ is the full nonlinear remainder. We use the heat kernel of Δ^E and retain the lower-order part of L_0 on the right-hand side. More precisely, set

$$\begin{aligned}\mathcal{D}_0 U := (L_0 - \Delta^E)U = &\sum_i R^N(U, e_i)e_i \\ &- 2\sum_{i,j}\langle \nabla^E_{X_i} U, e_j\rangle A_{ij}.\end{aligned} \tag{6.1}$$

Thus $\mathcal{D}_0 = B_0\nabla^E + C_0$ has uniformly bounded coefficients, and the equation becomes

$$(\partial_t - \Delta^E)U = \mathbf{H}_0 + \mathcal{D}_0 U + \mathcal{Q}(U), \qquad U(\cdot,0) = 0. \tag{6.2}$$

The coefficient bound for $\mathcal{D}_0$ uses only $|A_0|$ and $|\operatorname{Rm}_h|$; no derivative of A_0 is taken.

We record the time-derivative estimate for the heat potential explicitly. The distributional identity $\partial_t\mathcal{H}f = \operatorname{tr}_{g_0}\nabla^{E,2}\mathcal{H}f + f$ and Proposition 5.1 give

$$\begin{aligned}&\|\mathcal{H}f\|_{\mathcal{X}_\tau} + \|\partial_t\mathcal{H}f\|_{\mathcal{Y}_\tau} \le C_H\|f\|_{\mathcal{Y}_\tau}, \\ &\|\mathcal{H}f\|_{L^\infty(M\times(0,\tau))} \le C_H\sqrt{\tau}\,\|f\|_{\mathcal{Y}_\tau}, \qquad 0 < \tau \le T \le 1/4.\end{aligned} \tag{6.3}$$

Indeed, the weighted norm of the trace is at most $\sqrt{n}$ times the weighted norm of the Hessian. The second inequality is (5.16), restricted to $(0,\tau)$. We take $C_H \ge 1$. These estimates apply to measurable sources by the approximation argument in Proposition 5.1.

In particular, $\|\mathcal{H}f\|_{L^\infty(M\times(0,\tau))} \le C_H\sqrt{\tau}\,\|f\|_{\mathcal{Y}_T}$ proves that $\mathcal{H}f$ has zero uniform initial trace.

Lemma 6.1 (Weighted nonlinear estimates)**.** *There are constants $C_F, C_D, C_Q \ge 1$, depending only on n, m, such that the following holds. Let $0 < \tau \le 1/4$, and let $U, V \in \mathcal{X}_\tau$ satisfy*

$$m_U := \|U\|_{\mathcal{X}_\tau} \le \varepsilon_*, \qquad m_V := \|V\|_{\mathcal{X}_\tau} \le \varepsilon_*.$$

Then

$$\|\mathbf{H}_0\|_{\mathcal{Y}_\tau} \le C_F\sqrt{\tau}, \tag{6.4}$$

$$\|\mathcal{D}_0 U\|_{\mathcal{Y}_\tau} \le C_D\sqrt{\tau}\, m_U, \tag{6.5}$$

$$\|\mathcal{Q}(U)\|_{\mathcal{Y}_\tau} \le C_Q m_U^2, \tag{6.6}$$

$$\|\mathcal{Q}(U) - \mathcal{Q}(V)\|_{\mathcal{Y}_\tau} \le C_Q(m_U + m_V)\|U - V\|_{\mathcal{X}_\tau}. \tag{6.7}$$

Proof. The estimate for $\mathbf{H}_0$ follows from $|\mathbf{H}_0| \le \sqrt{n}$ and (5.4). Formula (6.1) gives $|\mathcal{D}_0 U| \le C(|U| + |\nabla^E U|)$, so the same bounded-to-weighted estimate supplies the factor $\sqrt{\tau}$ in (6.5).

For the nonlinear terms, put

$$z_U = |U| + |\nabla^E U|, \qquad h_U = |\nabla^{E,2} U|, \qquad d_j = |\nabla^{E,j}(U - V)| \quad (0 \le j \le 2).$$

For every bounded scalar function a and measurable tensor f, the definition on each admissible cylinder gives

$$\|af\|_{\mathcal{Y}_\tau} \le \|a\|_{L^\infty(M\times(0,\tau))}\|f\|_{\mathcal{Y}_\tau}. \tag{6.8}$$

Since $\|z_U\|_\infty \le m_U$ and $\|h_U\|_{\mathcal{Y}_\tau} \le m_U$, the exact one-section estimate (2.21) yields

$$\begin{aligned}\|\mathcal{Q}(U)\|_{\mathcal{Y}_\tau} &\le C\big(\|z_U^2\|_{\mathcal{Y}_\tau} + \|z_U h_U\|_{\mathcal{Y}_\tau}\big)\\ &\le C\big(\sqrt{\tau}\, m_U^2 + m_U\|h_U\|_{\mathcal{Y}_\tau}\big) \le C(1 + \sqrt{\tau})m_U^2.\end{aligned}$$

This proves (6.6) after enlarging C_Q.

The bound for $\nabla^{N,2}\operatorname{Rm}_h$ is used through the exact two-section estimate (2.24). Its three types of terms are estimated separately as follows:

$$\begin{aligned}\|(z_U + z_V)(d_0 + d_1)\|_{\mathcal{Y}_\tau} &\le \sqrt{\tau}\,(m_U + m_V)\|U - V\|_{\mathcal{X}_\tau},\\ \|(z_U + z_V)d_2\|_{\mathcal{Y}_\tau} &\le (m_U + m_V)\|\nabla^{E,2}(U - V)\|_{\mathcal{Y}_\tau},\\ \|(h_U + h_V)(d_0 + d_1)\|_{\mathcal{Y}_\tau} &\le \big(\|h_U\|_{\mathcal{Y}_\tau} + \|h_V\|_{\mathcal{Y}_\tau}\big)\|U - V\|_{\mathcal{X}_\tau}.\end{aligned}$$

The first line uses (5.4); the other two use (6.8). Since $\tau \le 1/4$, their sum proves (6.7) with a dimensional C_Q. These estimates concern the full remainder, including its dependence on the ambient curvature along the exponential curves. □

Define the map

$$\Phi(U) = \mathcal{H}\big(\mathbf{H}_0 + \mathcal{D}_0 U + \mathcal{Q}(U)\big). \tag{6.9}$$

It uses only the heat kernel K_E of Δ^E.

Proposition 6.2. *There are constants $\delta = \delta(n, m) > 0$ and $T_* = T_*(n, m) > 0$ such that Φ has a unique fixed point in the closed ball*

$$\mathbb{B}_{\delta,T_*} := \{U \in \mathcal{X}_{T_*} : \|U\|_{\mathcal{X}_{T_*}} \le \delta\}.$$

This fixed point solves (2.16)*, is smooth locally up to the initial time, and satisfies*

$$\|U\|_{\mathcal{X}_\tau} + \|\partial_t U\|_{\mathcal{Y}_\tau} \le C\sqrt{\tau}, \qquad \|U\|_{L^\infty(M\times[0,\tau])} \le C\tau, \qquad 0 < \tau \le T_*. \tag{6.10}$$

Proof. Step 1. Choice of the ball and the time interval. Fix the constants $C_H, C_F, C_D, C_Q \geq 1$ in (6.3) and Lemma 6.1. Choose

$$0 < \delta \leq \min\left\{\varepsilon_*, 1, \frac{1}{8C_H C_Q}\right\}, \tag{6.11}$$

and then choose $T_* > 0$ such that

$$\sqrt{T_*} \leq \min\left\{\frac{1}{2}, \frac{\delta}{4C_H C_F}, \frac{1}{4C_H C_D}\right\}. \tag{6.12}$$

For $U \in \mathbb{B}_{\delta,T_*}$, the pointwise jet is small almost everywhere, since $\| \, |U| + |\nabla^E U| \, \|_\infty \leq \delta$. The full remainder is therefore defined almost everywhere and belongs to $\mathcal{Y}_{T_*}$. The heat-potential estimate shows that $\Phi(U) \in \mathcal{X}_{T_*}$, including its zero uniform initial trace, and gives

$$\begin{aligned}\|\Phi(U)\|_{\mathcal{X}_{T_*}} &\leq C_H\big(C_F\sqrt{T_*} + C_D\sqrt{T_*}\,\delta + C_Q\delta^2\big)\\ &\leq \frac{\delta}{4} + \frac{\delta}{4} + \frac{\delta}{8} = \frac{5\delta}{8} < \delta.\end{aligned}$$

Thus Φ maps the closed ball into itself.

Step 2. Contraction and the exact equation. For U, V in this ball, linearity of $\mathcal{D}_0$ and (6.7) give

$$\begin{aligned}\|\Phi(U) - \Phi(V)\|_{\mathcal{X}_{T_*}} &\leq C_H\big(C_D\sqrt{T_*} + 2C_Q\delta\big)\|U - V\|_{\mathcal{X}_{T_*}}\\ &\leq \frac{1}{2}\|U - V\|_{\mathcal{X}_{T_*}}.\end{aligned}$$

The ball is complete because $\mathcal{X}_{T_*}$ is a Banach space. Starting with $U_0 = 0$ and setting $U_{j+1} = \Phi(U_j)$, the contraction estimate makes the successive differences summable in $\mathcal{X}_{T_*}$. The iterates converge in this norm to a fixed point U; the same contraction estimate proves uniqueness in the ball. The heat-potential identity implies (6.2) distributionally. Adding $\Delta^E U$ to $\mathcal{D}_0 U$ recovers $L_0 U$, so this is the exact equation (2.16), not an equation for a Taylor polynomial of its right-hand side. Equivalently, its coefficients are those in (2.17). On compact cylinders at positive times, the heat-potential identity gives $U \in W_p^{2,1}$; the local Sobolev chain rule therefore justifies the exponential-section identities for this weak solution. After Step 4 they hold classically.

Step 3. Estimates on every shorter time interval. Fix $0 < \tau \leq T_*$. Restriction decreases both weighted norms, and the heat potential at time $t \leq \tau$ uses only source values with $s < t$. Consequently, the same fixed-point identity holds on $(0, \tau)$. Put $m_\tau = \|U\|_{\mathcal{X}_\tau} \leq \delta$ and $M_\tau = m_\tau + \|\partial_t U\|_{\mathcal{Y}_\tau}$. The latter norm is finite by (6.3) applied to the fixed-point identity. The same estimate and Lemma 6.1 give

$$\begin{aligned}M_\tau &\leq C_H\big(C_F\sqrt{\tau} + C_D\sqrt{\tau}\,m_\tau + C_Q m_\tau^2\big)\\ &\leq C_H C_F\sqrt{\tau} + \left(\frac{1}{4} + \frac{1}{8}\right) m_\tau \leq C_H C_F\sqrt{\tau} + \frac{3}{8}M_\tau.\end{aligned}$$

Hence $M_\tau \leq (8/5)C_H C_F\sqrt{\tau}$. In particular,

$$\|\mathbf{H}_0 + \mathcal{D}_0 U + \mathcal{Q}(U)\|_{\mathcal{Y}_\tau} \leq C_F\sqrt{\tau} + C_D\sqrt{\tau}\,m_\tau + C_Q m_\tau^2 \leq C\sqrt{\tau}.$$

The extra $\sqrt{\tau}$ in the zeroth-order heat-potential estimate now gives $\|U\|_{L^\infty(M\times(0,\tau))} \leq C\tau$. This proves (6.10) for the fixed point just constructed; no new fixed point on a shorter interval is required.

Step 4. Local regularity at the initial time. The estimates just proved also supply unweighted local integrability up to $t = 0$. Indeed, with $r = \sqrt{s}$, the definition of $\mathcal{Y}_s$ gives

$$\int_{s/2}^{s} \int_{B(x,\sqrt{s})} (|\nabla^{E,2} U|^p + |\partial_t U|^p) \leq C s V_x(\sqrt{s}), \qquad 0 < s \leq T_*.$$

Fix a compact spatial set $D \subset M$. Cover it by balls of radius $\sqrt{s}$ with bounded overlap. Local volume comparison bounds the sum of their volumes by a constant times the volume of a fixed bounded neighborhood of D. Summing the preceding estimate over these balls and then over $s = 2^{-j} T_*$ yields

$$\int_0^{T_*} \int_D (|\nabla^{E,2} U|^p + |\partial_t U|^p) \leq C(D) \sum_{j=0}^{\infty} 2^{-j} T_* \leq 2C(D) T_*. \tag{6.13}$$

Together with the bounds for U and $\nabla^E U$, this is a local $W^{2,1}_p$ bound including the initial time. Covariant and coordinate derivatives are equivalent on each fixed compact coordinate patch. Since $p = n + 4 > n + 2$, parabolic Sobolev embedding gives a representative in $C^{1+\alpha,(1+\alpha)/2}_{\mathrm{loc}}$ for $0 < \alpha < 2/(n+4)$. Its value and spatial gradient at $t = 0$ vanish by (6.10).

In the exact quasilinear equation (2.17), the principal coefficient and the lower-order term are smooth functions of $x, U, \nabla^E U$ on every fixed coordinate patch. They are therefore locally parabolic C^α, and the principal coefficient is uniformly elliptic. Local linear parabolic regularity, applied to this equation with these coefficients, gives $C^{2+\alpha,1+\alpha/2}$ regularity on smaller cylinders, including those based at $t = 0$ because the initial section is identically zero. Repeating local regularity for the smooth quasilinear equation gives

$$U \in C^\infty(M \times (0, T_*]) \cap C^\infty_{\mathrm{loc}}(M \times [0, T_*]). \tag{6.14}$$

This last conclusion uses smoothness of the fixed initial data on compact sets. It imposes no uniform bound on derivatives of A_0: all constants used to choose δ, T_* were fixed in Steps 1–2, before this local regularity argument. Finally, Lemma 2.1 shows that $\widehat{F} = \mathrm{Exp}_{F_0} U$ is an immersion with metric uniformly equivalent to the complete metric g_0. It is therefore complete and solves the mean–DeTurck equation. □

The global bounds for U and $\nabla^E U$ come from the heat-potential estimate (6.3) and the restriction argument in Step 3. The weighted Hessian and time-derivative bounds give local $W^{2,1}_p$ control through (6.13), which is used for local regularity. Sobolev embedding at this stage does not give an L^∞ bound for the Hessian. To obtain that bound uniformly, we apply local L^p estimates first to the rescaled equation and then to its spatial derivative, and finally use Sobolev embedding. The initial bound for A_0 supplies the required normalisation; no global curvature bound for the evolving immersion is assumed in this argument.

Lemma 6.3. *Let U be the solution constructed above and let $\widehat{F} = F_U$. After decreasing $T_* > 0$ by a dimensional amount, we have*

$$\sup_{M \times [0,T_*]} \big(|A_{\widehat{F}}| + |W(\widehat{F})|_{g_0} + |\nabla^{E,2} U| + |\partial_t U| \big) \leq C(n, m). \tag{6.15}$$

In particular, the constant does not depend on the injectivity radii of M or N, or on higher derivatives of A_0.

Proof. The bounds already established in (6.10) give

$$\|U\|_{L^\infty(M\times(0,s))} \le Cs, \qquad \|\nabla^E U\|_{L^\infty(M\times(0,s))} \le C\sqrt{s}. \tag{6.16}$$

Their proof used neither a pointwise bound for $\nabla^{E,2}U$ nor curvature bounds for the evolving immersion. We use them to estimate the coordinate equation for $\widehat{F}$ below.

Step 1. Uniform local coordinates for the immersion. Fix $x_0 \in M$ and let $y_0 = F_0(x_0)$. The bounds for Rm_{g_0} and Rm_h, together with the harmonic-coordinate construction of Lemma 3.4, provide harmonic coordinates on fixed-radius exponential pullback balls about x_0 and y_0. Take the source harmonic coordinates with a uniform $W^{2,2p}$ bound, and the target coordinates with a uniform $W^{2,q}$ bound, where $q > \max\{n+m, 2p\}$ is fixed in terms of the dimensions. For a sufficiently small fixed $\beta > 0$, Sobolev embedding gives uniform $C^{1,\beta}$ bounds for both metrics, in addition to uniform ellipticity. All radii and constants depend only on n and m. These coordinates are on the pullback balls; no injectivity-radius bound on the original manifolds is required.

The additional ambient bound $|\nabla^N \mathrm{Rm}_h| \le 1$ gives one further coordinate derivative of the target metric. In target harmonic coordinates,

$$h^{ab}D_aD_bh_{\lambda\mu} = -2\,\mathrm{Ric}(h)_{\lambda\mu} + \mathcal{R}_{\lambda\mu}(h, Dh), \tag{6.17}$$

where $\mathcal{R}$ is quadratic in Dh, with smooth coefficients depending on h and h^{-1}. The coordinate derivative of $\mathrm{Ric}(h)$ is bounded, because it is the sum of its covariant derivative and contractions of $\Gamma(h)$ with $\mathrm{Ric}(h)$. Differentiating (6.17) once gives

$$\begin{aligned} h^{ab}D_aD_b(D_\gamma h_{\lambda\mu}) = &-2D_\gamma\,\mathrm{Ric}(h)_{\lambda\mu} - (D_\gamma h^{ab})D_aD_bh_{\lambda\mu} \\ &+ D_\gamma\mathcal{R}_{\lambda\mu}(h, Dh). \end{aligned}$$

The right-hand side has a uniform L^q bound: Dh is bounded and $D^2h \in L^q$. Interior elliptic $W^{2,q}$ estimates for $D_\gamma h$, on a fixed smaller target ball, yield $\|h\|_{W^{3,q}} \le C$. Since $q > n+m$, Sobolev embedding gives $|D^2h| \le C$, and hence

$$\|\Gamma(h)\|_{L^\infty} + \|D\Gamma(h)\|_{L^\infty} \le C. \tag{6.18}$$

For the source metric we need only

$$\|\Gamma(g_0)\|_{L^\infty} + \|D\Gamma(g_0)\|_{L^p} \le C, \tag{6.19}$$

which follows from its harmonic-coordinate $W^{2,2p}$ bound. In particular, no bound for $\nabla\,\mathrm{Rm}_{g_0}$ or ∇A_0 is used.

After shrinking the source ball by a dimensional factor, the lifted initial immersion takes values in the target coordinate ball. To see this directly, lift the curves $s \mapsto F_0(\exp^{g_0}_{x_0}(sv))$ through the target exponential map, starting at the origin. Their lengths are $|v|$. Since the target pullback ball has a fixed distance from its centre to its boundary, these lifts exist for all the source vectors under consideration. Local invertibility of the exponential map and smooth dependence of the lift on v give a smooth lifted immersion. The curves $s \mapsto \mathrm{Exp}_{F_0(x)}(sU(x,t))$ lift from its endpoints in the same way. By (6.16), their lengths are at most Ct, so the lifted maps $\widehat{F}_t$ remain in the target coordinate ball for all sufficiently small T_*.

In these coordinates, the components of F_0 satisfy

$$\partial_i\partial_jF_0^\alpha = \Gamma(g_0)^k_{ij}\partial_kF_0^\alpha - \Gamma(h)^\alpha_{\lambda\mu}(F_0)\partial_iF_0^\lambda\partial_jF_0^\mu + (A_0)^\alpha_{ij}. \tag{6.20}$$

Metric equivalence, the isometry of F_0, and the bound for A_0 therefore imply $|DF_0| + |D^2F_0| \le C$ on a fixed smaller source ball. Here D denotes ordinary coordinate differentiation. The Jacobi-field estimate (2.10) and the bounded target Christoffel symbols convert (6.16) into

$$|\widehat{F} - F_0| \le Ct, \qquad |D\widehat{F} - DF_0| \le C\sqrt{t}. \tag{6.21}$$

In the second estimate, parallel transport and comparison of coordinate components differ by at most $C|U|$.

Step 2. A rescaled interior estimate. Fix $t_0 \in (0, T_*]$ and put $r = \sqrt{t_0}/8$. Translate the source coordinates so that $x_0 = 0$, and define

$$\ell(x) = F_0(0) + DF_0(0)x, \qquad v(z,\tau) = r^{-2}\big(\widehat{F}(rz, t_0 + r^2\tau) - \ell(rz)\big)$$

on $\mathscr{Q}_2 = B_2(0) \times (-4, 0]$. The choice of r ensures that the corresponding time interval stays strictly above zero. Equations (6.20) and (6.21) give

$$|v| + |D_z v| \le C \qquad \text{on } \mathscr{Q}_2. \tag{6.22}$$

The coordinate equation (2.13) becomes

$$\partial_\tau v^\alpha = a^{ij}\partial_i\partial_j v^\alpha + B^\alpha, \tag{6.23}$$

where, evaluated at $(x, s) = (rz, t_0 + r^2\tau)$,

$$a^{ij} = g(\widehat{F})^{ij}, \qquad B^\alpha = a^{ij}\left[-\Gamma(g_0)^k_{ij}\partial_k\widehat{F}^\alpha + \Gamma(h)^\alpha_{\lambda\mu}(\widehat{F})\partial_i\widehat{F}^\lambda\partial_j\widehat{F}^\mu\right].$$

There is no scaling factor in front of B, because $\widehat{F} - \ell$ was divided by r^2. Let $a_0^{ij} = g_0^{ij}(0)$. The coefficients satisfy

$$\|a - a_0\|_{L^\infty(\mathscr{Q}_2)} \le Cr, \qquad \|B\|_{L^\infty(\mathscr{Q}_2)} \le C, \tag{6.24}$$

and a_0 is uniformly positive definite.

Choose a smooth cutoff η supported in $\mathscr{Q}_2$, equal to one on $\mathscr{Q}_{3/2}$, and vanishing near the bottom time and the spatial boundary. Apply the constant-coefficient parabolic L^{2p} estimate to ηv, extended by zero in space. Its equation has the form

$$(\partial_\tau - a_0^{ij}\partial_i\partial_j)(\eta v) = (a - a_0)^{ij}\partial_i\partial_j(\eta v) + G, \qquad \|G\|_{L^{2p}} \le C,$$

where the commutators contain only v, Dv, and derivatives of η. The use of $2p$ here is legitimate because these terms and B are bounded; it does not require an L^{2p} hypothesis in the original weighted space. Moreover, ηv is smooth on a compact cylinder at positive times by (6.14). Its $W^{2,1}_{2p}$ norm is therefore finite before applying the estimate; no uniform bound for this norm is being assumed. By (6.24), the first term on the right can be absorbed after decreasing T_* so that the constant in the L^{2p} estimate times Cr is smaller than $1/2$. We obtain

$$\|v\|_{W^{2,1}_{2p}(\mathscr{Q}_{3/2})} \le C. \tag{6.25}$$

We differentiate the coordinate equation (6.23) once in a spatial variable. Its coefficients involve h, $\Gamma(h)$, and $\Gamma(g_0)$; the required derivatives are controlled by (6.18) and (6.19). This step does not differentiate the exponential-section remainder $\mathcal{Q}$, whose coefficients already involve $\nabla^N \mathrm{Rm}_h$. Put $w_k = \partial_k v$. It satisfies

$$\partial_\tau w_k - a^{ij}\partial_i\partial_j w_k = G_k := (\partial_k a^{ij})\partial_i\partial_j v + \partial_k B. \tag{6.26}$$

To estimate G_k, write

$$F_r(z,\tau) = \widehat{F}(rz, t_0 + r^2\tau) = \ell(rz) + r^2 v(z,\tau),$$
$$J_r(z,\tau) = D_x\widehat{F}(rz, t_0 + r^2\tau) = DF_0(0) + rD_z v.$$

The coefficient a is a smooth function of $h(F_r)$ and J_r on a uniformly elliptic set. Since $D_z F_r = rJ_r$ and $D_z J_r = rD_z^2 v$, we have

$$|D_z a| \le Cr(1 + |D_z^2 v|). \tag{6.27}$$

The formula for B, together with (6.18), similarly implies

$$|D_z B| \le Cr\big(1 + |D_z^2 v| + |D_x\Gamma(g_0)|(rz)\big). \tag{6.28}$$

The last source coefficient is independent of time. Therefore (6.19) gives

$$\|rD_x\Gamma(g_0)(r\,\cdot)\|_{L^p(\mathscr{Q}_{3/2})} \le Cr^{1-n/p}\|D_x\Gamma(g_0)\|_{L^p(B_{3r/2})} \le C, \tag{6.29}$$

because $p = n+4 > n$ and $r \le 1$. By (6.25), the quadratic term in the Hessian belongs to L^p. More precisely,

$$\|(D_z a) * D_z^2 v\|_{L^p(\mathscr{Q}_{3/2})} \le Cr\big(\|D_z^2 v\|_{L^p} + \|D_z^2 v\|_{L^{2p}}^2\big) \le C.$$

Equations (6.28) and (6.29) thus imply $\|G_k\|_{L^p(\mathscr{Q}_{3/2})} \le C$.

Apply a second cutoff, supported in $\mathscr{Q}_{3/2}$ and equal to one on $\mathscr{Q}_1$, to (6.26). The resulting commutators contain w_k and $Dw_k = D\partial_k v$, which are controlled respectively by (6.22) and (6.25). The constant-coefficient L^p estimate and the same absorption argument give

$$\|w_k\|_{W_p^{2,1}(\mathscr{Q}_1)} \le C, \qquad k = 1, \dots, n. \tag{6.30}$$

Since $p > n+2$, parabolic Sobolev embedding applied to w_k controls its first spatial derivatives pointwise. Consequently,

$$\|D_z^2 v\|_{L^\infty(\mathscr{Q}_{1/2})} \le C. \tag{6.31}$$

The equation (6.23) and the boundedness of a, B give $\|\partial_\tau v\|_{L^\infty(\mathscr{Q}_{1/2})} \le C$ as well. In particular,

$$|D_z^2 v(0,0)| + |\partial_\tau v(0,0)| \le C.$$

Since $D_x^2\widehat{F} = D_z^2 v$ and $\partial_t\widehat{F} = \partial_\tau v$, scaling back gives

$$|D_x^2\widehat{F}(x_0,t_0)| + |\partial_t\widehat{F}(x_0,t_0)| \le C. \tag{6.32}$$

This estimate is uniform in x_0 and t_0. Only estimates on compact coordinate cylinders at positive times were used in deriving it. The constants in the two absorption arguments depend on the uniform coordinate bounds and (6.16). Local higher-derivative bounds supply only the finiteness needed to apply these estimates and do not enter their constants or the choice of T_*.

Step 3. Geometric consequences. Taking the normal part of the covariant coordinate Hessian in (6.32) gives $|A_{\widehat{F}}| \le C$. Furthermore, the first derivatives of the induced metric are bounded in these coordinates, since they involve only $D\widehat{F}$, $D^2\widehat{F}$, and the first derivatives of h. It follows from (2.11) that $|W(\widehat{F})|_{g_0} \le C$.

Finally, differentiate $\widehat{F} = \mathcal{E}(U)$ twice covariantly. The term containing $\nabla^{E,2}U$ is $P_U\mathcal{V}_U(\nabla^{E,2}U)$; all remaining terms are bounded by Lemma 2.1, $|A_0| \le 1$, and the bounds

for U and $\nabla^E U$. Since $\mathcal{V}_U^{-1}$ is uniformly bounded, we obtain $|\nabla^{E,2}U| \le C$. The identity $P_U^{-1}\partial_t \widehat{F} = \mathcal{V}_U(\partial_t U)$ gives $|\partial_t U| \le C$ as well. The initial-time bounds follow from $U(\cdot,0) = 0$, the smoothness already proved in (6.14), and $|A_0| \le 1$. This proves (6.15). □

Under the scaling $\widetilde{h} = \rho^{-2}h$, the preceding construction has time interval $0 \le \widetilde{t} \le T_*(n,m)$ and $|A|^2_{\widetilde{h}} \le C(n,m)$. Returning to h gives $0 \le t \le \rho^2 T_*(n,m)$ and

$$|A_t|_h^2 \le C(n,m)(\Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2}),$$

where $\rho^{-2} = \Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2}$ when this quantity is positive. If it vanishes, $F_t = F_0$ is a stationary solution.

6.2. **Removal of the DeTurck gauge.** We now remove the gauge using the pointwise estimates of Lemma 6.3. We first work in the normalised scale; the original scale is restored at the end of the construction.

Proof of Theorem 1.1: removal of the gauge. Let

$$F_u(\cdot,t) : M \longrightarrow N, \qquad 0 \le t \le T,$$

be the mean–DeTurck flow obtained in Proposition 6.2, and set

$$g(t) := F_u(t)^* h.$$

By construction, F_u satisfies

$$\partial_t F_u = \mathbf{H}(F_u) + dF_u(W_t), \tag{6.33}$$

where $W_t \in \Gamma(TM)$ is the DeTurck vector field

$$W_t^k = g^{ij}(t)\big(\Gamma(g(t))_{ij}^k - \Gamma(g_0)_{ij}^k\big). \tag{6.34}$$

Equivalently,

$$\big(\partial_t F_u\big)^{\perp} = \mathbf{H}(F_u),$$

and the tangential component is given by

$$\big(\partial_t F_u\big)^{\top} = dF_u(W_t).$$

We first collect the estimates needed to integrate the time-dependent vector field W_t. By Lemma 2.1 and the smallness of U, the induced metrics are uniformly equivalent:

$$C^{-1}g_0 \le g(t) \le Cg_0 \qquad \text{on } M \times [0,T]. \tag{6.35}$$

The first derivatives of $g(t)$ involve $\nabla^{E,2}U$. Since the difference of two Levi-Civita connections is a tensor, we have

$$\Gamma(g(t)) - \Gamma(g_0) = \frac{1}{2} g(t)^{-1} * \nabla^{g_0} g(t), \tag{6.36}$$

and hence, by (6.34),

$$|W_t|_{g_0} \le C|\nabla^{g_0} g(t)|_{g_0}. \tag{6.37}$$

Here, C depends only on m and n.

Lemma 6.3 gives

$$\sup_{x\in M} |W_t(x)|_{g_0} \le C, \qquad 0 \le t \le T. \tag{6.38}$$

Consequently,

$$\int_0^T \|W_t\|_{L^\infty(M,g_0)}\,\mathrm{d}t \le CT. \tag{6.39}$$

We next solve the time-dependent ordinary differential equation

$$\frac{d}{dt}\varphi_t(x) = -W_t\big(\varphi_t(x)\big), \qquad \varphi_0(x) = x. \tag{6.40}$$

For each $x \in M$, local existence and uniqueness follow from the local smoothness of W_t up to $t = 0$, established in (6.14). To prove that the solution exists on the whole interval $[0, T]$, let

$$\gamma(t) := \varphi_t(x).$$

Then, with respect to the complete metric g_0, we have

$$d_{g_0}\big(\gamma(t), x\big) \le \int_0^t |\dot\gamma(s)|_{g_0}\,ds = \int_0^t |W_s(\gamma(s))|_{g_0}\,ds \le \int_0^t |W_s|_{L^\infty(M,g_0)}\,ds.$$

By (6.39), the right-hand side is finite uniformly for $t \le T$. Thus every integral curve remains in a bounded g_0-ball on each finite time interval. Since (M, g_0) is complete, Hopf–Rinow implies that closed bounded g_0-balls are compact. The standard continuation theorem for ordinary differential equations therefore extends $\varphi_t(x)$ to all $t \in [0, T]$.

We now show that φ_t is a diffeomorphism of M for every $t \in [0, T]$. Fix $t_0 \in [0, T]$ and consider the backward equation

$$\frac{d}{ds}\psi_s(x) = W_{t_0-s}\big(\psi_s(x)\big), \qquad \psi_0(x) = x, \qquad 0 \le s \le t_0. \tag{6.41}$$

The same estimate as above yields existence on the whole interval $[0, t_0]$. By uniqueness for the time-dependent ODE,

$$\psi_{t_0} \circ \varphi_{t_0} = \mathrm{Id}_M, \qquad \varphi_{t_0} \circ \psi_{t_0} = \mathrm{Id}_M\,.$$

Hence φ_{t_0} is a diffeomorphism, with inverse ψ_{t_0}.

Define

$$\widetilde{F}(x,t) := F_u\big(\varphi_t(x), t\big). \tag{6.42}$$

We claim that $\widetilde{F}$ satisfies the mean curvature flow equation. By the chain rule and (6.42),

$$\begin{aligned}\partial_t\widetilde{F}(x,t) &= \partial_t F_u\big(\varphi_t(x),t\big) + dF_u\big|_{(\varphi_t(x),t)}\big(\partial_t\varphi_t(x)\big)\\ &= \partial_t F_u\big(\varphi_t(x),t\big) - dF_u\big|_{(\varphi_t(x),t)}\big(W_t(\varphi_t(x))\big).\end{aligned}$$

Using (6.33), we obtain

$$\partial_t\widetilde{F}(x,t) = \mathbf{H}(F_u)\big(\varphi_t(x),t\big). \tag{6.43}$$

It remains to identify the right-hand side with the mean curvature vector of $\widetilde{F}$. Since

$$\widetilde{F}_t = F_u(t) \circ \varphi_t,$$

the induced metric of $\widetilde{F}_t$ is

$$\widetilde{g}(t) := \widetilde{F}_t^* h = \varphi_t^* g(t).$$

The normal bundle and the second fundamental form are natural under reparametrization. More precisely,

$$A_{\widetilde{F}_t} = \varphi_t^* A_{F_u(t)},$$

that is,

$$A_{\widetilde{F}_t}(X,Y) = A_{F_u(t)}\big(d\varphi_t(X), d\varphi_t(Y)\big) \circ \varphi_t.$$

Taking the trace with respect to $\widetilde{g}(t) = \varphi_t^* g(t)$, we obtain

$$\mathbf{H}(\widetilde{F}_t) = \mathcal{H}(F_u(t)) \circ \varphi_t. \tag{6.44}$$

Combining (6.43) and (6.44) gives

$$\partial_t \widetilde{F} = \mathbf{H}(\widetilde{F}).$$

Furthermore,

$$\widetilde{F}(\cdot,0) = F_u(\varphi_0(\cdot),0) = F_u(\cdot,0) = F_0.$$

Thus $\widetilde{F}$ is a mean curvature flow with initial immersion F_0.

Finally, all geometric estimates obtained for the mean–DeTurck flow are preserved under reparametrization. Indeed,

$$\widetilde{g}(t) = \varphi_t^* g(t), \qquad A_{\widetilde{F}_t} = \varphi_t^* A_{F_u(t)},$$

and therefore

$$|A_{\widetilde{F}_t}|_{\widetilde{g}(t)} = |A_{F_u(t)}|_{g(t)} \circ \varphi_t.$$

Therefore,

$$\sup_{M\times[0,T]} |A_{\widetilde{F}_t}|^2 = \sup_{M\times[0,T]} |A_{F_u(t)}|^2,$$

and the curvature bound of Lemma 6.3, after scaling back, gives

$$\sup_{M\times[0,T]} |A_{\widetilde{F}_t}|^2 \le C(n,m)\big(\Lambda^2 + K_0 + K_1^{2/3} + K_2^{1/2}\big).$$

Moreover, $(M, g(t))$ is complete by (6.35), and $(M, \widetilde{g}(t))$ is isometric to it under φ_t. Thus the reparametrised time slices are complete. Together with the existence construction above, this proves Theorem 1.1. □

7. Uniqueness of mean curvature flow

This section concerns uniqueness without a positive lower bound for the injectivity radius of the ambient manifold. The argument uses a common harmonic-map gauge and a comparison on the fixed pullback bundle.

Let

$$F_a : M^n \times [0,T] \longrightarrow (\overline{M}^{n+m}, \overline{g}), \qquad a = 1,2,$$

be two complete smooth mean curvature flows satisfying

$$\partial_t F_a = \mathbf{H}_a, \qquad F_a(\cdot,0) = F_0. \tag{7.1}$$

Assume

$$\sup_{\overline{M}} \Big(|\overline{\mathrm{Rm}}| + |\overline{\nabla \mathrm{Rm}}| + |\overline{\nabla}^2 \overline{\mathrm{Rm}}| \Big) \le K \tag{7.2}$$

and

$$\sup_{M\times[0,T]} |A_a| \le A_*, \qquad a = 1,2. \tag{7.3}$$

Write

$$g_a(t) = F_a(t)^* \overline{g}, \qquad g_0 = F_0^* \overline{g}.$$

We shall use the spaces introduced in Section 5. For completeness, fix

$$p = n + 4 > n + 2$$

and, for

$$0 < t \le T \le 1, \qquad 0 < r \le \min\{1, \sqrt{t}\},$$

define

$$Q_r(x,t) := B_{g_0}(x,r) \times (t - r^2/2, t). \tag{7.4}$$

If $V_x(r) = \mathrm{Vol}_{g_0}(B_{g_0}(x,r))$, set

$$\|f\|_{Y_T} := \sup_{x,t,r} r \left(\frac{1}{r^2 V_x(r)} \int_{Q_r(x,t)} |f|^p \, d\mu_{g_0} \, ds \right)^{1/p}, \tag{7.5}$$

and, for a section u with $u(\cdot, 0) = 0$,

$$\begin{aligned} \|u\|_{X_T} := {} & \|u\|_{L^\infty(M\times(0,T))} + \|\nabla^E u\|_{L^\infty(M\times(0,T))} \\ & + \sup_{x,t,r} r \left(\frac{1}{r^2 V_x(r)} \int_{Q_r(x,t)} |\nabla^{E,2} u|^p \, d\mu_{g_0} \, ds \right)^{1/p}. \end{aligned} \tag{7.6}$$

We use only the heat potential for the fixed rough Laplacian:

$$\mathcal{H}_E f(t) = \int_0^t e^{(t-s)\Delta^E} f(s) \, ds.$$

Proposition 5.1, after scaling if necessary, gives

$$\|\mathcal{H}_E f\|_{X_T} \le C_E \|f\|_{Y_T}. \tag{7.7}$$

Every first-order and zeroth-order term of the linearized equation is kept on its right-hand side. Thus the proof does not use a heat kernel for the linearization containing A_0, or any derivative of its first-order coefficients.

Under (7.2), Lemma 2.4 gives the pointwise bound (2.24) for the full remainder $\mathcal{Q}$. For the equation with the common fixed reference metric h chosen below, we denote the entire nonconstant right-hand side by $\mathcal{N}$. It includes the bounded first-order and zeroth-order linear terms. Step 5 proves that, for

$$B_{\delta,T} := \{u \in X_T : \|u\|_{X_T} \le \delta\},$$

one has

$$\|\mathcal{N}(u) - \mathcal{N}(v)\|_{Y_T} \le C_\theta(\delta + \sqrt{T}) \|u - v\|_{X_T} \qquad (u, v \in B_{\delta,T}). \tag{7.8}$$

Indeed, the Hessian terms in (2.24) are controlled by the last summand of the X_T norm. Bounded first-order and zeroth-order terms instead acquire a factor $\sqrt{T}$ through (5.4). The effect of the change of reference metric is computed explicitly in Step 5.

Theorem 7.1. *Under assumptions* (7.2)–(7.3)*,*

$$F_1 = F_2 \qquad \text{on } M \times [0,T].$$

No positive lower bound for $\mathrm{inj}(\overline{M}, \overline{g})$ *is required.*

Proof. We divide the proof into eight steps.

Step 1: estimates for the induced metrics.

The metric evolution equation is

$$\partial_t (g_a)_{ij} = -2\langle \mathbf{H}_a, A^{(a)}_{ij} \rangle_{\overline{g}}. \tag{7.9}$$

Since $|\mathbf{H}_a| \le \sqrt{n}\,|A_a|$,

$$|\partial_t g_a|_{g_a} \le C(n) A_*^2. \tag{7.10}$$

Consequently,

$$e^{-C(n)A_*^2 t} g_0 \le g_a(t) \le e^{C(n)A_*^2 t} g_0. \tag{7.11}$$

In particular, every $g_a(t)$ is complete.

The Gauss equation gives

$$\mathrm{Rm}(g_a) = F_a^* \overline{\mathrm{Rm}} + A_a * A_a, \tag{7.12}$$

and hence

$$|\mathrm{Rm}(g_a)| \le C(n)(K + A_*^2). \tag{7.13}$$

For clarity, the first-derivative estimate does not require a global bound for ∇A_a at the initial time. With $\mathscr{L}_a = \partial_t - \Delta_{g_a(t)}$, the curvature evolution and its first covariant derivative give

$$\mathscr{L}_a |A_a|^2 \le -|\nabla A_a|^2 + C, \qquad \mathscr{L}_a |\nabla A_a|^2 \le -|\nabla^2 A_a|^2 + C|\nabla A_a|^2 + C.$$

The second ambient curvature derivative in (7.2) controls the differentiated forcing. To localize, use nonincreasing distance cutoffs $\chi_R = \zeta(d_{g_a(t)}(o,\cdot)/R)^2$, $R \ge 1$. Laplacian comparison and (7.10) give

$$|\nabla \chi_R| \le C R^{-1}, \qquad \mathscr{L}_a \chi_R \le C(1 + R^{-1} + R^{-2})$$

in the usual barrier interpretation at the cut locus. The negative gradient term in the evolution of $L|A_a|^2$ absorbs the cutoff terms for $t\chi_R^2 |\nabla A_a|^2 + L|A_a|^2$, with L large. The compact-domain maximum principle and then $R \to \infty$ give

$$\sqrt{t}\, |\nabla A_a| \le C \qquad \text{on } M \times (0, T_1], \tag{7.14}$$

where $T_1 > 0$ and C depend only on n, m, K, A_*.

Differentiating (7.9) and (7.12), we obtain

$$\sqrt{t}\, (|\nabla \partial_t g_a| + |\nabla \mathrm{Rm}(g_a)|) \le C. \tag{7.15}$$

Step 2: choice of a common regular target metric.

The fixed target in the harmonic-map gauge must have bounded curvature derivative. This need not hold uniformly for g_0, because differentiating the Gauss equation for g_0 produces the term

$$A_0 * \nabla A_0.$$

We therefore follow the positive-time target construction.

Fix

$$0 < \theta < \min\{T, T_1\}$$

and set

$$h := g_1(\theta). \tag{7.16}$$

By (7.11),

$$C^{-1} g_0 \le h \le C g_0,$$

so (M, h) is complete. By (7.12) and (7.14),

$$|\mathrm{Rm}(h)|_h + |\nabla^h \mathrm{Rm}(h)|_h \le C_\theta. \tag{7.17}$$

The constant is allowed to depend on the fixed positive number θ. Unless otherwise stated, constants in the remainder of the uniqueness proof may also depend on θ.

Both harmonic-map flows below use this same target metric h.

The identity map also has bounded second covariant derivative. Indeed, if $S_a = \partial_t g_a$, then

$$\partial_t \Gamma(g_a)^k_{ij} = \frac{1}{2} g_a^{k\ell}\big(\nabla_i (S_a)_{j\ell} + \nabla_j (S_a)_{i\ell} - \nabla_\ell (S_a)_{ij}\big).$$

Metric equivalence and (7.15) therefore give

$$|\nabla^{g_a(t)} - \nabla^{g_0}|_{g_0} \le C \int_0^t s^{-1/2}\, ds \le C\sqrt{t}. \tag{7.18}$$

In particular,

$$|\nabla d\,\mathrm{Id}\,|_{g_0,h} = |\Gamma(h) - \Gamma(g_0)|_{g_0,h} \le C\sqrt{\theta}. \tag{7.19}$$

Step 3: harmonic-map gauges.

For $a = 1, 2$, solve

$$\partial_t \phi_a = \tau_{g_a(t),h}(\phi_a), \qquad \phi_a(\cdot, 0) = \mathrm{Id}_M\,. \tag{7.20}$$

We apply Huang–Tam's short-time existence theorem [6, Theorem 1.1]. Its hypotheses can be checked as follows:

(1) By (7.13) and (7.10),

$$2\,\mathrm{Ric}(g_a(t)) + \partial_t g_a(t) \ge -C g_a(t).$$

(2) Equations (7.10) and (7.15) imply

$$|\partial_t g_a| \le C, \qquad |\nabla \partial_t g_a| \le C t^{-1/2}.$$

These bounds are stronger than the corresponding time-weighted hypotheses in Huang–Tam.

(3) The target (M, h) is complete and has bounded sectional curvature.

(4) The energy density of the initial identity map

$$\mathrm{Id} : (M, g_a(0)) = (M, g_0) \longrightarrow (M, h)$$

is uniformly bounded. The same bound holds with domain metric $g_a(t)$, by (7.11).

(5) The terminal domain metric is complete and has bounded curvature by (7.13). It therefore admits the smooth distance-like exhaustion function with bounded gradient and Hessian required in [6, Theorem 1.1]; see the discussion immediately following that theorem.

Thus (7.20) has a smooth solution on a common interval $[0, T_2]$.

The Bochner formula and the generalized maximum principle give

$$|d\phi_a|_{g_a(t),h} \le C. \tag{7.21}$$

The equation for

$$\Psi_a = \nabla d\phi_a$$

contains the terms

$$\nabla\,\mathrm{Rm}(g_a) * d\phi_a, \qquad \nabla \partial_t g_a * d\phi_a, \qquad \nabla\,\mathrm{Rm}(h) * (d\phi_a)^{*4}.$$

Here $(d\phi_a)^{*4}$ denotes a contraction of four factors of $d\phi_a$. Set

$$P_a = |\Psi_a|^2, \qquad e_a = |d\phi_a|^2.$$

The differentiated equation and the Bochner formula imply

$$(\partial_t - \Delta_{g_a})P_a \le -2|\nabla\Psi_a|^2 + C_\theta P_a + C_\theta(1+t^{-1}), \tag{7.22}$$

$$(\partial_t - \Delta_{g_a})e_a \le -2P_a + C_\theta. \tag{7.23}$$

The t^{-1} term in the first inequality comes from applying Young's inequality to $Ct^{-1/2}|\Psi_a|$. Before that application, the Kato inequality has the stronger form

$$(\partial_t - \Delta_{g_a})|\Psi_a| \le C_\theta|\Psi_a| + C_\theta(1+t^{-1/2}) \tag{7.24}$$

in the barrier sense.

We justify the global comparison in two stages. Since h is complete and has bounded curvature, choose a smooth proper distance-like function $\rho \ge 1$ with $|d\rho|_h + |\nabla^{h,2}\rho|_h \le C_\theta$. Equation (7.18) gives uniform bounds for the gradient and Hessian of ρ with respect to $g_a(t)$. Choose $\zeta \in C^\infty([0,\infty))$, equal to one on $[0,1]$, equal to zero on $[2,\infty)$, and taking values in $[0,1]$, and put $\chi_R = \zeta(\rho/R)^2$, $R \ge 1$. Then

$$|\nabla\chi_R| \le C_\theta R^{-1}, \qquad |\Delta_{g_a}\chi_R| \le C_\theta(R^{-1}+R^{-2}), \qquad \frac{|\nabla\chi_R|^2}{\chi_R} \le C_\theta R^{-2} \tag{7.25}$$

where the quotient is evaluated on $\{\chi_R > 0\}$.

First consider

$$Q_R = t\chi_R^2 P_a + Le_a.$$

Using (7.22) and (7.23), the cross term is bounded by

$$2t|\langle\nabla(\chi_R^2), \nabla P_a\rangle| \le t\chi_R^2|\nabla\Psi_a|^2 + Ct|\nabla\chi_R|^2 P_a.$$

Thus, for L sufficiently large and $t \le \min\{T_2, 1\}$, with constants independent of R,

$$(\partial_t - \Delta_{g_a})Q_R \le C_\theta.$$

On the parabolic boundary of a compact domain containing the support of χ_R, $Q_R \le L\sup e_a$; the same bound holds at $t = 0$. The ordinary maximum principle on that domain therefore gives $t\chi_R^2 P_a \le C_\theta$. Letting $R \to \infty$, we obtain

$$|\Psi_a(x,t)| \le C_\theta t^{-1/2}.$$

Only local smoothness at the initial time is used in this first stage.

For the second stage, regularize the norm by $f_\delta = (P_a + \delta^2)^{1/2}$, $0 < \delta \le 1$, and set $q_R = \chi_R f_\delta$. At a positive spatial maximum of q_R, $\nabla f_\delta = -f_\delta\nabla\chi_R/\chi_R$. Applying the regularized version of (7.24) gives

$$(\partial_t - \Delta_{g_a})q_R \le C_\theta q_R + C_\theta(1+t^{-1/2}) + f_\delta\left(-\Delta_{g_a}\chi_R + 2\frac{|\nabla\chi_R|^2}{\chi_R}\right).$$

The rough estimate already proved and (7.25) bound the last term above by

$$C_\theta(R^{-1}+R^{-2})(t^{-1/2}+\delta).$$

All terms on the right are integrable in time. Moreover, (7.19) bounds $q_R(\cdot,0)$ independently of R and δ. Comparison with the corresponding scalar ODE on the compact support of q_R gives

$$\sup_M q_R(\cdot,t) \le e^{C_\theta t}\left[C_\theta + \delta + C_\theta(t+\sqrt{t}) + C_\theta(R^{-1}+R^{-2})(\sqrt{t}+\delta t)\right].$$

Letting $R \to \infty$ and then $\delta \downarrow 0$, we conclude that

$$|\nabla d\phi_a| \le C_\theta \qquad \text{on } M \times [0, T_2]. \tag{7.26}$$

This argument uses neither a uniform bound for higher derivatives of the initial immersion nor the heat kernel of a fixed pullback bundle.

Let

$$\widetilde{g}_a = \phi_a^* h.$$

The tensor evolution equation for $\widetilde{g}_a$, together with (7.21) and (7.26), gives

$$(\partial_t - \Delta_{g_a})\widetilde{g}_a \ge -C_\theta(\widetilde{g}_a + g_a)$$

in the sense of symmetric two-tensors. Since

$$\widetilde{g}_a(0) = h \ge C^{-1} g_0 = C^{-1} g_a(0),$$

the noncompact tensor maximum principle yields, after decreasing T_2,

$$C^{-1} g_a(t) \le \phi_a(t)^* h \le C g_a(t). \tag{7.27}$$

Hence $\phi_a(t)$ is a local diffeomorphism.

Moreover, Huang–Tam's tension-field estimate gives

$$|\partial_t \phi_a| \le C t^{-1/2},$$

and therefore

$$d_h(\phi_a(x,t), x) \le \int_0^t |\partial_s \phi_a(x,s)|\, ds \le C\sqrt{t}. \tag{7.28}$$

Thus $\phi_a(t)$ is proper. A proper local diffeomorphism is a covering map. Since the homotopy

$$s \longmapsto \phi_a(\cdot, s), \qquad 0 \le s \le t,$$

joins $\phi_a(t)$ to the identity, its induced map on the fundamental group is surjective, after identifying basepoints along the homotopy. The covering therefore has one sheet. Consequently,

$$\phi_a(t) : M \longrightarrow M \tag{7.29}$$

is a global diffeomorphism.

Set

$$\psi_a = \phi_a^{-1}, \qquad \widehat{F}_a = F_a \circ \psi_a, \qquad \widehat{g}_a = \widehat{F}_a^* \overline{g}.$$

Then

$$\widehat{F}_1(\cdot, 0) = \widehat{F}_2(\cdot, 0) = F_0.$$

Define

$$W^k(\widehat{g}, h) = \widehat{g}^{ij}\big(\Gamma_{ij}^k(\widehat{g}) - \Gamma_{ij}^k(h)\big). \tag{7.30}$$

The inverse-gauge calculation gives

$$\partial_t \widehat{F}_a = \mathbf{H}(\widehat{F}_a) + d\widehat{F}_a\big(W(\widehat{g}_a, h)\big), \tag{7.31}$$

or equivalently

$$\partial_t \widehat{F}_a = \mathrm{tr}_{\widehat{g}_a} \nabla^{h,\overline{g}} d\widehat{F}_a. \tag{7.32}$$

The chain rule, (7.18), and the two-sided differential bounds for ϕ_a and its inverse give $|\nabla^{g_0,\overline{g}} d\widehat{F}_a| \le C$. The strictly parabolic equation then bounds $\partial_t \widehat{F}_a$. Thus

$$\begin{aligned} &C^{-1} g_0 \le \widehat{g}_a(t) \le C g_0, \\ &|\partial_t \widehat{F}_a| \le C, \\ &|\nabla^{g_0,\overline{g}} d\widehat{F}_a| \le C. \end{aligned} \tag{7.33}$$

Step 4: lifting through the exponential map.

The ambient sectional-curvature upper bound gives a uniform lower bound for the conjugate radius. Choose

$$0 < r_0 < \frac{\pi}{2\sqrt{K_+}}, \qquad K_+ = \max\{K, 1\}. \tag{7.34}$$

For every $p \in \overline{M}$ and $|v| \le r_0$, the vertical differential

$$D_v \exp_p : T_p\overline{M} \longrightarrow T_{\exp_p(v)}\overline{M}$$

is invertible with a uniform inverse bound.

Let

$$E = F_0^* T\overline{M}, \qquad \mathcal{E}(x, v) = \exp_{F_0(x)} v.$$

Lemma 7.2. *After decreasing T_2, there is a unique smooth section $u_a \in \Gamma(E)$ such that*

$$\widehat{F}_a(x,t) = \mathcal{E}(x, u_a(x,t)), \qquad u_a(\cdot, 0) = 0, \qquad |u_a(x,t)| < r_0. \tag{7.35}$$

Proof. Fix $x \in M$. Since $D_v\mathcal{E}(x, 0)$ is the identity, the inverse function theorem produces a lift of $t \mapsto \widehat{F}_a(x,t)$ for sufficiently small time.

Once the identity

$$\mathcal{E}(x, u_a(x,t)) = \widehat{F}_a(x,t)$$

has been established, it may be differentiated to give

$$D_v\mathcal{E}(x, u_a)\, \partial_t u_a = \partial_t \widehat{F}_a. \tag{7.36}$$

This equation is not used to define the lift before (7.35) is known.

The inverse bound for $D_v\mathcal{E}$ and (7.33) imply

$$|u_a(x,t)| \le Ct.$$

Thus, for small T_2, the lift remains in $\{|v| < r_0\}$. A maximal lift cannot terminate earlier: it has finite length, converges to a vector in the fiber ball, and the inverse function theorem extends it.

Local uniqueness follows from local invertibility of $v \mapsto \mathcal{E}(x, v)$. Smooth dependence on x follows from the parameter-dependent inverse function theorem. □

This construction uses only the unique lift of the particular path starting at the zero vector. It does not require $\exp_{F_0(x)}$ to be globally injective on a ball.

Step 5: the fixed-bundle equation.

Let $\nabla^E = F_0^* \overline{\nabla}$. We write the equation for $\widehat{F} = \mathcal{E}(u)$ using Δ^E as its only linear operator on the left-hand side. We first compute precisely the effect of the common reference metric h.

Set

$$C_{ij}^k = \Gamma(g_0)_{ij}^k - \Gamma(h)_{ij}^k.$$

Equation (7.18) gives $|C|_{g_0} \le C_\theta$. Relative to the equation with reference metric g_0, the immersion equation acquires the term

$$g(\widehat{F})^{ij} C^k_{ij} \partial_k \widehat{F}.$$

After radial parallel transport to E_x and multiplication by the inverse vertical differential, this is exactly

$$\mathcal{B}_\theta(u, Z) = \mathcal{M}^{ij}_u C^k_{ij} E^u_k(1), \qquad Z = \nabla^E u, \qquad \mathcal{M}^{ij}_u = g(F_u)^{ij} \mathcal{V}_u^{-1}.$$

Here $E^u_k(1) = P_u^{-1} dF_u(X_k)$ and all quantities are those used in Lemma 2.4. At a fixed point, use a g_0-orthonormal frame and put

$$a^{ij}_u = \langle Z_i, e_j \rangle + \langle e_i, Z_j \rangle, \qquad P^u_k = E^u_k(1) - e_k - Z_k, \qquad \mathcal{D}^{ij}_u = \mathcal{M}^{ij}_u - \delta^{ij} \operatorname{Id} + a^{ij}_u \operatorname{Id}.$$

The exact decomposition is

$$\mathcal{B}_\theta(u, Z) = \mathcal{B}_\theta(0, 0) + \mathcal{D}_\theta u + \mathcal{R}_\theta(u, Z),$$

where

$$\begin{aligned} \mathcal{B}_\theta(0,0) &= \delta^{ij} C^k_{ij} e_k, \\ \mathcal{D}_\theta u &= \delta^{ij} C^k_{ij} Z_k - a^{ij}_u C^k_{ij} e_k, \\ \mathcal{R}_\theta(u, Z) &= C^k_{ij} \big[\delta^{ij} P^u_k - a^{ij}_u Z_k - a^{ij}_u P^u_k + \mathcal{D}^{ij}_u (e_k + Z_k + P^u_k) \big]. \end{aligned}$$

Thus $\mathcal{D}_\theta$ has order one, with bounded coefficients; no derivative of the connection difference C is involved. The exact Jacobi and inverse-matrix estimates in Lemma 2.4 give

$$\begin{aligned} |P^u| + |\mathcal{D}_u| &\le C z_u^2, \\ |P^u - P^v| + |\mathcal{D}_u - \mathcal{D}_v| &\le C(z_u + z_v)(|u - v| + |\nabla^E(u - v)|), \end{aligned}$$

where $z_u = |u| + |\nabla^E u|$. Splitting each product in the displayed formula for $\mathcal{R}_\theta$ therefore gives

$$|\mathcal{R}_\theta(u) - \mathcal{R}_\theta(v)| \le C_\theta (z_u + z_v)(|u - v| + |\nabla^E(u - v)|). \tag{7.37}$$

For reference metric g_0, define the bounded lower-order operator

$$\mathcal{D}_0 w = \sum_i \overline{R}(w, e_i) e_i - 2 \sum_{i,j} \langle \nabla^E_{X_i} w, e_j \rangle A_{ij}.$$

The full fixed-bundle equation is consequently

$$\partial_t u - \Delta^E u = P_0 + \mathcal{N}(u), \qquad u(\cdot, 0) = 0, \tag{7.38}$$

where

$$P_0 = \mathbf{H}_0 + \mathcal{B}_\theta(0, 0), \qquad |P_0| \le C_\theta,$$

and

$$\mathcal{N}(u) = \mathcal{D}_0 u + \mathcal{D}_\theta u + \mathcal{Q}(u) + \mathcal{R}_\theta(u, \nabla^E u). \tag{7.39}$$

The only second-derivative coefficient on the right is

$$\mathcal{A}(u, \nabla^E u) = g(F_u)^{-1} - g_0^{-1}, \qquad \mathcal{A}(0, 0) = 0. \tag{7.40}$$

In particular, the change of reference metric adds no second-order term. Quantitatively, for every $w \in X_T$,

$$\|(\mathcal{D}_0 + \mathcal{D}_\theta) w\|_{Y_T} \le C_\theta \sqrt{T}\, \|w\|_{X_T}. \tag{7.41}$$

For $u, v \in B_{\delta,T}$, (7.37) gives

$$\|\mathcal{R}_\theta(u) - \mathcal{R}_\theta(v)\|_{Y_T} \le C_\theta \delta \sqrt{T}\, \|u - v\|_{X_T}.$$

Combining these inequalities with (2.24) proves (7.8). All constants depend on the fixed geometric bounds and θ, not on an ambient injectivity radius.

Once membership in X_T has been established in Step 6, the right-hand side of (7.38) belongs to Y_T, and its heat potential is well defined. The mild equation is

$$u = \mathcal{H}_E\big(P_0 + \mathcal{N}(u)\big). \tag{7.42}$$

Indeed, the difference between the two sides is a bounded solution of the homogeneous connection heat equation with zero uniform initial trace. The Kato inequality and the scalar maximum principle on the complete manifold (M, g_0), whose Ricci curvature is bounded below, show that this difference is zero. We do not construct or estimate a heat kernel for $\Delta^E + \mathcal{D}_0 + \mathcal{D}_\theta$.

Step 6: admissibility of arbitrary DeTurck solutions.

To compare the two given flows, we first show that their lifts belong to a small X_T-ball.

Lemma 7.3. *There is $T_3 > 0$ such that*

$$\|u_a\|_{X_{T_3}} \le C\sqrt{T_3}, \qquad a = 1, 2. \tag{7.43}$$

In particular, after decreasing T_3,

$$u_1, u_2 \in B_{\delta,T_3}.$$

Proof. We use the geometric bounds before applying any estimate in X_T. The path-lifting argument gives $|u_a| \le Ct$. Differentiating $\widehat{F}_a = \mathcal{E}(u_a)$ once, using the uniform bounds for $d\widehat{F}_a$, and inverting the vertical differential gives $|\nabla^E u_a| \le C$. Differentiating a second time and using the Jacobi-field bounds gives

$$|\nabla^{E,2} u_a| \le C\big(1 + |\nabla^{g_0,\bar{g}} d\widehat{F}_a| + |\nabla^E u_a|^2\big) \le C.$$

These are finite, pointwise estimates, independent of any prior membership in X_T.

To improve the first derivative bound, fix a g_0-unit vector X at x, follow the complete g_0-geodesic with initial velocity X, and parallel transport the section back to E_x. Taylor's formula along that geodesic gives, for $0 < \ell \le 1$,

$$|\nabla^E_X u_a(x,t)| \le \frac{2\|u_a(\cdot,t)\|_\infty}{\ell} + \frac{\ell}{2}\|\nabla^{E,2} u_a(\cdot,t)\|_\infty.$$

Taking $\ell = \sqrt{t}$ yields

$$|u_a| + |\nabla^E u_a| \le C\sqrt{t}. \tag{7.44}$$

Finally, the pointwise Hessian bound implies

$$\sup_{x,t,r} r \left(\frac{1}{r^2 V_x(r)} \int_{Q_r(x,t)} |\nabla^{E,2} u_a|^p \, d\mu_{g_0}\, ds \right)^{1/p} \le C\sqrt{T}.$$

Together with (7.44), this proves (7.43). The proof uses (7.26) through (7.33). □

This lemma supplies the step that cannot be obtained merely from the general estimate $\sqrt{t}\,|\nabla d\phi_a| \le C$.

Step 7: comparison in X_T.

Both u_1 and u_2 satisfy (7.42). The common forcing P_0 cancels upon subtraction, and hence

$$u_1 - u_2 = \mathcal{H}_E\big(\mathcal{N}(u_1) - \mathcal{N}(u_2)\big).$$

Using (7.7) and (7.8), we obtain

$$\begin{aligned}\|u_1 - u_2\|_{X_T} &\le C_E\|\mathcal{N}(u_1) - \mathcal{N}(u_2)\|_{Y_T}\\ &\le C_E C_\theta(\delta + \sqrt{T})\|u_1 - u_2\|_{X_T}.\end{aligned}$$

Choose first δ, and then T, so that

$$C_E C_\theta(\delta + \sqrt{T}) < \frac{1}{2}.$$

It follows that

$$u_1 = u_2.$$

Consequently,

$$\widehat{F}_1 = \widehat{F}_2 =: \widehat{F}. \tag{7.45}$$

Step 8: recovery of the original flows and continuation.

Let

$$\widehat{g} = \widehat{F}^* \overline{g}$$

and let $W = W(\widehat{g}, h)$ be defined by (7.30). With this sign convention, the common equation for ϕ_a is

$$\partial_t \phi_a = -W(t) \circ \phi_a, \qquad \phi_a(\cdot, 0) = \mathrm{Id}_M, \tag{7.46}$$

The inverse maps satisfy the corresponding transport equation

$$\partial_t \psi_a = d\psi_a(W);$$

This identity follows by differentiating $\phi_a \circ \psi_a = \mathrm{Id}$.

Since $\widehat{F}$ is smooth, W is locally Lipschitz in space on every compact subset of $M \times [0, T_3]$. Ordinary differential equation uniqueness applied pointwise to (7.46) gives

$$\phi_1 = \phi_2.$$

Together with

$$F_a = \widehat{F}_a \circ \phi_a$$

and (7.45), this proves

$$F_1 = F_2$$

on a nontrivial interval beginning at $t = 0$.

Finally, let

$$\mathcal{I} := \{t \in [0, T] : F_1 = F_2 \text{ on } M \times [0, t]\}.$$

The set $\mathcal{I}$ is nonempty and closed. If $t_0 \in \mathcal{I}$ and $t_0 < T$, then the two flows have the same immersion and the same induced metric at t_0. The bounds on the second fundamental form and the ambient curvature derivatives persist on the remaining interval. The same first-derivative estimate can therefore be applied after a time shift. Reapplying the argument to

$$\widetilde{F}_a(x, s) = F_a(x, t_0 + s)$$

extends equality beyond t_0. Hence $\mathcal{I}$ is open in $[0, T]$. Since $[0, T]$ is connected,

$$\mathcal{I} = [0, T].$$

Therefore

$$F_1 = F_2 \qquad \text{on } M \times [0,T].$$

□

Appendix A. Harmonic map heat flow

We record the Hessian estimates used in the uniqueness argument. Let $\phi : (M, g(t)) \to (N, h)$ be a smooth harmonic map heat flow,

$$\partial_t \phi = \tau_{g(t),h}(\phi), \tag{A.1}$$

and write $S = \partial_t g$, $e = |d\phi|^2$, and $\Psi = \nabla d\phi$. The Bochner formula is

$$(\partial_t - \Delta_g)e = -2|\Psi|^2 - (2\operatorname{Ric}(g) + S)(d\phi, d\phi) + 2\mathcal{R}_h(d\phi, d\phi, d\phi, d\phi). \tag{A.2}$$

In particular, the hypotheses

$$2\operatorname{Ric}(g) + S \geq -\Lambda g, \tag{A.3}$$

and

$$\sec_h \leq \kappa \tag{A.4}$$

give

$$(\partial_t - \Delta_g)e \leq \Lambda e + C\kappa_+ e^2. \tag{A.5}$$

For the solution constructed by [6, Theorem 1.1], bounded initial energy and the geometric hypotheses of that theorem yield a uniform short-time energy bound. The proposition below assumes this bound explicitly and supplies the full Hessian estimates.

Proposition A.1. *Suppose $0 < T \leq 1$ and $g(t)$ is a smooth complete family. Assume that there is a smooth proper function $\rho : M \to [1, \infty)$ such that*

$$|\nabla^{g(t)}\rho| + |\Delta_{g(t)}\rho| \leq C_0.$$

Suppose also that

$$|\operatorname{Rm}(g)| + |S| + |\operatorname{Rm}(h)| + |\nabla \operatorname{Rm}(h)| \leq C_0, \qquad |\nabla \operatorname{Rm}(g)| + |\nabla S| \leq C_0(1 + t^{-1/2}), \tag{A.6}$$

and that the smooth solution of (A.1) *satisfies*

$$\sup_{M \times [0,T]} |d\phi| \leq E_0. \tag{A.7}$$

Then

$$\sqrt{t}\,|\nabla d\phi| \leq C.$$

If in addition $\sup_M |\nabla d\phi(\cdot, 0)| \leq B_0$, then

$$\sup_{M \times [0,T]} |\nabla d\phi| \leq C(1 + B_0).$$

The constants depend only on n, $\dim N$, C_0, and E_0.

Proof. Twice differentiating the map equation and commuting derivatives gives

$$\begin{aligned}(D_t - \Delta_g)\Psi = {} & \operatorname{Rm}(g) * \Psi + \nabla \operatorname{Rm}(g) * d\phi + \operatorname{Rm}(h) * (d\phi)^{*2} * \Psi \\ & + \nabla \operatorname{Rm}(h) * (d\phi)^{*4} + \nabla S * d\phi + S * \Psi.\end{aligned}$$

Here D_t is the covariant time derivative; the terms involving S also include those obtained by differentiating the tensor norm. Writing $P = |\Psi|^2$, we obtain

$$(\partial_t - \Delta_g)P \leq -2|\nabla \Psi|^2 + CP + C(1 + t^{-1}), \tag{A.8}$$

and the Bochner formula gives

$$(\partial_t - \Delta_g)e \le -2P + C. \tag{A.9}$$

Before applying Young's inequality to the forcing term, the Kato inequality reads

$$(\partial_t - \Delta_g)|\Psi| \le C|\Psi| + C(1 + t^{-1/2}).$$

Choose $\zeta \in C^\infty([0,\infty))$, equal to one on $[0,1]$ and zero on $[2,\infty)$, with $0 \le \zeta \le 1$, and set $\chi_R = \zeta(\rho/R)^2$ for $R \ge 1$. Then

$$|\nabla\chi_R| \le CR^{-1}, \qquad |\Delta_g\chi_R| \le C(R^{-1} + R^{-2}), \qquad \frac{|\nabla\chi_R|^2}{\chi_R} \le CR^{-2}$$

on $\{\chi_R > 0\}$. Define

$$Q_R = t\chi_R^2 P + Le. \tag{A.10}$$

The cross term in its evolution satisfies

$$2t|\langle\nabla\chi_R^2, \nabla P\rangle| \le t\chi_R^2|\nabla\Psi|^2 + Ct|\nabla\chi_R|^2 P.$$

Thus (A.8) and (A.9) imply $(\partial_t - \Delta_g)Q_R \le C$ when L is sufficiently large, uniformly in R. On the lateral boundary of a compact domain containing $\operatorname{supp}\chi_R$, and at the initial time, $Q_R \le LE_0^2$. The ordinary compact-domain maximum principle gives $t\chi_R^2 P \le C$. Sending $R \to \infty$ proves $\sqrt{t}\,|\Psi| \le C$.

For the second conclusion, let $f_\delta = (P + \delta^2)^{1/2}$ and $q_R = \chi_R f_\delta$, where $0 < \delta \le 1$. At a positive spatial maximum of q_R, the regularized Kato inequality gives

$$(\partial_t - \Delta_g)q_R \le Cq_R + C(1 + t^{-1/2}) + f_\delta\left(-\Delta_g\chi_R + 2\frac{|\nabla\chi_R|^2}{\chi_R}\right).$$

By the first conclusion, the last term is bounded above by $C(R^{-1} + R^{-2})(t^{-1/2} + \delta)$. Comparing with a scalar ODE on the compact support, and using $q_R(\cdot,0) \le B_0 + \delta$, yields

$$\sup_M q_R(\cdot,t) \le e^{Ct}\left[B_0 + \delta + C(t + \sqrt{t}) + C(R^{-1} + R^{-2})(\sqrt{t} + \delta t)\right].$$

Letting $R \to \infty$ and then $\delta \downarrow 0$ proves the result. All comparisons have taken place on compact domains, so no growth assumption on the Hessian at spatial infinity has been used. □

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